

IMPERIAL

DEVELOPING AN ASSISTIVE LOW-COST MOBILE BIMANUAL MANIPULATOR

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Abstract

Assistive robots have the potential to improve independence and quality of life for individuals with motor impairments by supporting activities of daily living. However, existing mobile bimanual manipulation platforms capable of conducting research in this area are often restrictively expensive and require substantial technical knowledge to operate. These barriers limit accessibility and hinder the development of user-centred assistive robotic systems.

This thesis presents the Resource-efficient Intelligent General-purpose mobile Bimanual robotic sYstem (RIGBY), an accessible end-to-end platform for mobile bimanual manipulation. Building upon the XLeRobot platform, RIGBY introduces a number of hardware and software extensions, including an asymmetric bimanual manipulator configuration, a whole-body teleoperation system for demonstration collection, and a multimodal user interface supporting graphical, textual, and voice interaction. To improve usability throughout the learning pipeline, the interface additionally provides tools for task selection, dataset and calibration management, demonstration recording, training configuration, policy execution, and perception-assisted task discovery through object inspection.

To support learning from demonstration, the system adopts the Action Chunking with Transformers (ACT) framework and explores a novel extension termed Dominant-hand Action Chunking with Transformers (DOM-ACT). DOM-ACT incorporates an explicit dominant-support structure intended to better model the asymmetric roles commonly observed during bimanual manipulation tasks. A complete URDF model of the platform was additionally developed to facilitate future simulation-based research.

Experimental evaluation demonstrated that RIGBY was capable of learning and executing a range of mobile bimanual manipulation tasks using low-cost hardware. Comparative experiments between ACT and DOM-ACT suggested that introducing task-relevant inductive biases may improve performance on tasks exhibiting dominant-support characteristics. However, the evaluation also highlighted limitations associated with low-cost actuators, environment-specific imitation learning, and assumptions regarding fixed manipulator roles.

Overall, this work demonstrates that meaningful research into mobile bimanual manipulation

can be conducted using substantially lower-cost hardware than many existing platforms. In addition to this, it highlights that accessibility in assistive robotics depends not only on affordability, but also on making these systems easier to interact with, teach, and use.

Plain Language Summary

Robots that can assist people with everyday tasks have the potential to improve independence and quality of life, particularly for individuals living with disabilities. However, many research robots capable of performing these tasks are extremely expensive, making them difficult to access outside well-funded laboratories.

This project explored whether a more affordable robot could still learn useful household skills. To investigate this, a low-cost mobile robot called RIGBY was developed. RIGBY can move around the home, use two robotic arms to manipulate objects, and learn new tasks by watching demonstrations provided through a remote control system. As opposed to relying solely on traditional menus and controls, RIGBY also supports voice commands, a touchscreen interface, and the ability to recognise objects in its environment to suggest tasks it already knows how to perform.

The project additionally investigated a new method for teaching robots with two arms. Many everyday activities involve one hand performing the main action while the other provides support. A new learning approach, called Dominant Arm Action Chunking Transformer DOM-ACT, was proposed to reflect this behaviour and compared against an existing imitation learning method.

The results showed that a low-cost robot could successfully learn and perform a range of bimanual tasks, including activities requiring coordination between movement and manipulation. At the same time, the project highlighted challenges that remain before such systems can become practical household assistants, including hardware reliability, the need for demonstration data, and difficulties adapting to new environments.

Overall, this work demonstrates that accessible assistive robotics is not only about reducing cost, but also about making robots easier to teach, easier to interact with, and more practical for the people who may ultimately benefit from them.

Declaration of Originality

I hereby declare that the work presented in this thesis is my own unless otherwise stated. To the best of my knowledge the work is original and ideas developed in collaboration with others have been appropriately referenced. The work I am submitting represents my independent effort.

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List of Acronyms

ADL Activity of Daily Living

HRI Human-Robot Interaction

IADL Instrumental Activity of Daily Living

RIGBY Resource-efficient Intelligent General-purpose mobile Bimanual robotic sYstem

ACT Action Chunking with Transformers

DOM-ACT Dominant-hand Action Chunking with Transformers

URDF Unified Robot Description Format

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1

Introduction

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Research has shown that young children with motor impairments are at risk of developing a reduced sense of self-efficacy as they feel they lack control over their environment [1]. This lack of agency can lead to decreased curiosity, early passivity, and increased dependence on caregivers, potentially resulting in learned helplessness and reduced participation in developmental activities [2].

Evidence suggests that independent interaction with the environment, particularly self-initiated movement, plays a critical role in early psychological, cognitive, and social development [2]. Studies involving laboratory-designed power mobility devices and modified robotic platforms have demonstrated improvements in motivation, task mastery, and social engagement among children with motor impairments [2]. Early exposure to power mobility allows children to experience the consequences of their actions, supporting a transition from passive observation to active exploration.

Assistive robotics therefore offers potential benefits beyond functional task support. By enabling individuals with motor impairments to exert influence over their environment, assistive robotic systems may counteract learned helplessness and improve self-perceived competence [1]. Supporting participation in Activity of Daily Livings (ADLs) is specifically important, as reduced engagement can contribute to further deterioration of skills and increasing social isolation [2]. Robot assistance that enables continued participation may help bridge this “skill-learning gap”.

Despite this potential, much existing research in assistive robotics has centred on adult users, clinical rehabilitation, or high-cost systems that are unsuitable for long-term use in environments such as home or school [3]. Furthermore, many advanced mobile manipulation platforms capable of supporting assistive robotics research remain restrictively expensive, limiting their accessibility to homes and educational institutions. Existing systems also frequently require substantial technical expertise to operate, creating barriers to the collection of demonstrations, training of learning policies, and deployment of robotic behaviours [4].

Imitation learning has emerged as a promising paradigm for developing assistive robotic behaviours, allowing robots to acquire skills directly from demonstrations as opposed to through explicit programming. Recent methods such as Action Chunking with Transformers (ACT) have demonstrated strong performance on manipulation tasks using more affordable robotic platforms than those traditionally employed within manipulation research [5]. Nevertheless, many of these systems remain excessively expensive, and thus unlikely to be economically accessible for deployment within homes, schools, or community settings. Comparatively little work has explored the application of imitation learning approaches to genuinely low-cost mobile bimanual platforms within the context of assistive robotics.

Moreover, while open-source projects have helped reduce the financial barriers associated with robotics research, many still require substantial technical expertise to assemble, calibrate, teach, and operate. The supporting infrastructure required to make these systems practical—including demonstration interfaces, training tools, and intuitive mechanisms for human–robot interaction—is therefore often overlooked.

In response to these challenges, this project investigates whether a low-cost, modular mobile bimanual robotic platform can provide an accessible foundation for assistive robotics research and deployment based on learning from demonstration. To explore this question, the Resource-efficient Intelligent General-purpose mobile Bimanual robotic sYstem RIGBY was developed. Building upon the XLeRobot platform [6], RIGBY integrates whole-body teleoperation for demonstration collection, imitation learning for skill acquisition, and multimodal interaction capabilities intended to simplify the processes through which users and researchers teach, discover, and utilise robotic behaviours.

The primary objective of this work is therefore to develop and evaluate an accessible mobile bimanual manipulation platform that supports imitation learning research while remaining aligned with the long-term goals of assistive robotics. In this context, accessibility is considered not only

in terms of hardware affordability, but also in reducing the barriers associated with teaching, interacting with, and deploying robotic systems. By making these technologies easier to use and investigate, this work aims to contribute towards the broader development of assistive robots capable of supporting independence and participation in everyday life.

1.1 Contributions of this Work

The principal contributions of this project are summarised below:

1. **Development of an accessible mobile bimanual manipulation platform.** A low-cost robotic platform was developed by extending the XLeRobot architecture with an asymmetric bimanual configuration, onboard sensing, mobile manipulation capabilities, and a whole-body teleoperation system for demonstration collection.
2. **Introduction of Dominant-hand Action Chunking with Transformers (DOM-ACT).** A novel extension to the ACT framework was proposed to explicitly model dominant–support relationships commonly observed in asymmetric bimanual tasks. Comparative experiments were conducted to investigate its potential advantages over conventional ACT policies.
3. **Development of a multimodal interaction framework for assistive robotic systems.** The RIGBY interface integrates touchscreen controls, textual interaction, voice commands, task cataloguing, dataset management, training utilities, and object inspection tools supporting policy selection and system diagnostics, thereby improving accessibility throughout the robot teaching process.
4. **Creation of a reusable simulation model for future research.** A complete URDF representation of the RIGBY platform was developed to facilitate simulation-based experimentation and future extensions of the system.
5. **Empirical evaluation of low-cost mobile bimanual imitation learning.** The project provides evidence that meaningful research into mobile bimanual manipulation can be conducted using substantially lower-cost hardware while also highlighting practical limitations relating to actuator performance, environment specificity, and demonstration requirements.

1.2 Report Structure

The remainder of this report is organised as follows. Chapter 2 reviews relevant literature on assistive robotics, bimanual manipulation, imitation learning, and existing robotic platforms. Chapter 3 outlines the system requirements and project objectives. Chapter 4 presents the analysis and overall system design. Chapters 5 and 6 describe the hardware and software implementation of RIGBY respectively. Chapter 7 details the evaluation methodology adopted throughout this work, while Chapter 8 presents the experimental results. Chapter 9 discusses the findings in relation to the original objectives and considers limitations of the proposed approach. Finally, Chapter 10 concludes the report and outlines directions for future work.

2

Background

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Assistive robotics has emerged as an important research area that aims to support people with physical, cognitive, or sensory impairments in performing ADLs [3]. Advances in robotic manipulation, sensing, and Human-Robot Interaction (HRI) have enabled systems capable of providing functional assistance in domestic, educational, and care-related environments. Despite this progress, many assistive robotic systems remain expensive, complex, and difficult to adapt to the specific needs of individual users [7].

This chapter reviews literature relevant to the design of a low-cost bimanual robotic assistant for users with motor impairments. Key topics include assistive robotics, the functional impact

of upper-limb motor impairments, bimanual robotic manipulation, existing assistive solutions, imitation-learning platforms, and low-cost modular robotic systems. Through this review, limitations of current approaches are identified, motivating the development of an accessible mobile bimanual assistive robotic system.

2.1 Assistive Robotics

2.1.1 Definition and Scope

Assistive robotics is a subfield of robotics concerned with developing systems that support individuals affected by physical, cognitive, or sensory impairments in performing ADLs and Instrumental Activity of Daily Livings (IADLs) [8]. It differs from fully autonomous service robotics because assistive robots typically involve the user in the control loop, providing support while preserving user agency over task execution [9]. The central objective is therefore not to replace human caregivers or maximise autonomy, but to enhance independence, participation, and quality of life.

Within this field, the present project focuses on robotic systems that support task-oriented manipulation as opposed to long-term therapeutic or clinical outcomes. The scope is limited to non-clinical assistance for everyday manipulation tasks, with emphasis on affordability, usability, safety, and accessibility. This definition provides the foundation for examining bimanual robotic systems, low-cost mobile manipulation platforms, and learning-based approaches in the following sections.

2.1.2 Assistive Robotics for Users with Motor Impairments

The design of assistive robots must account for substantial variation between users. Motor impairments can affect limb strength, coordination, range of motion, fatigue, and the ability to reach, grasp, or manipulate objects. These limitations can reduce independence in tasks such as eating, dressing, writing, tidying, and organising personal spaces, often increasing reliance on caregivers [10]. Variability between individuals, and even between limbs of the same individual, highlights the need for flexible assistive solutions that can adapt to different physical capabilities.

Many ADLs and IADLs require coordinated use of both hands. In these tasks, one limb often stabilises an object while the other performs a more precise action [11]. Limitations in bimanual

coordination can therefore significantly restrict functional independence. Assistive robotic systems that support bimanual interaction offer the potential to complement a user’s residual abilities by providing stabilisation, coordinated manipulation, or task-level assistance without removing the user entirely from the activity.

Although pediatric assistive contexts provide an important motivation for this work, the system developed in this project is evaluated as a general low-cost assistive manipulation platform. The focus is therefore on representative assistive manipulation behaviours as opposed to claims specific to any single user group.

2.2 Bimanual Robotics in Assistive Applications

Bimanual robotic manipulation involves the coordinated use of two robotic arms to interact with objects or the environment [11]. Compared to single-arm manipulation, bimanual systems enable handling of larger or more complex objects and support tasks that require stabilisation and coordinated motion. These capabilities are in particular relevant for everyday tasks, for example, cleaning surfaces, opening containers or organising objects.

Despite these advantages, bimanual manipulation introduces increased system complexity, including inter-arm coordination, collision avoidance, and force regulation [11]. These challenges are amplified in assistive contexts, where safety, compliance, and robustness are critical. As a result, many assistive robotic systems rely on single-arm designs, which are simpler and more affordable but limited in their ability to support bimanual tasks [9].

Dual-arm assistive robots offer greater functional capability by enabling coordinated manipulation without relying on the user’s physical stabilisation. However, these systems are less common due to cost and control complexity [11]. This motivates investigation into whether learning-based control and task-level abstractions can enable effective bimanual assistance using low-cost hardware.

2.3 Existing Assistive Robotic Systems

A variety of assistive robotic systems have been developed to support individuals with upper-limb motor impairments. These systems differ in mechanical design, autonomy, cost, and intended

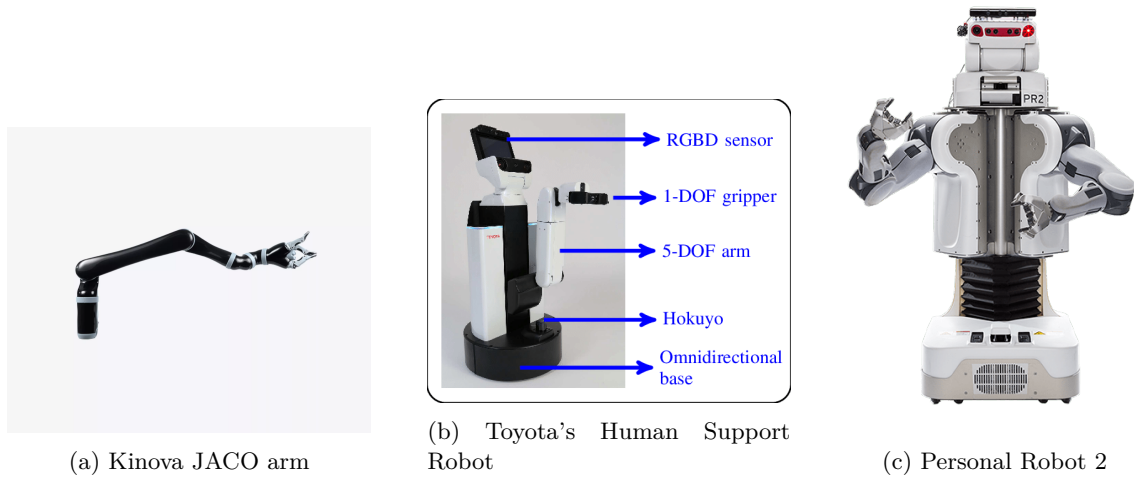


Figure 2.1: Single-arm assistive manipulators demonstrate practical reaching and grasping support, but also highlight the limitations of one-handed assistance for bimanual activities of daily living.

deployment context. While many demonstrate strong performance in specific areas, each class of system presents limitations when considered for accessible bimanual assistance in everyday environments.

2.3.1 Single-Arm Assistive Manipulators

Single-arm assistive robots, for instance as the Kinova JACO arm, are among the most widely deployed assistive robotic systems for daily tasks including feeding, object retrieval, and simple manipulation [12]. These systems benefit from reliable hardware, mature control software, and well-established user interfaces, making them suitable for domestic and clinical environments [12].

However, the use of a single manipulator inherently limits the range of tasks that can be performed. Many ADLs and IADLs require stabilisation and coordinated bimanual interaction, which single-arm systems cannot provide without additional environmental support [9]. Furthermore, the high cost and proprietary nature of many commercial platforms restrict accessibility and adaptability, especially for long-term use outside clinical settings.

2.3.2 Mobile Assistive Robots

Mobile assistive robots, for example Toyota's Human Support Robot (HSR), integrate autonomous navigation and manipulation to support household assistance tasks [13]. These platforms demonstrate the feasibility of mobile manipulation in unstructured indoor environments and highlight

the advantages of combining mobility with assistive manipulation [14].

Despite these strengths, most mobile assistive robots employ only a single manipulator, limiting their effectiveness for tasks requiring coordinated bimanual action [14]. In addition, these platforms are typically expensive and complex, and are often designed for adult users or research environments [14]. This reduces their suitability for deployment in home or educational contexts, in particular for pediatric users.

2.3.3 Research-Oriented Bimanual Robotic Platforms

Research platforms such as the PR2 robot and other dual-arm humanoid systems demonstrate advanced bimanual manipulation capabilities, including coordinated grasping, force sharing, and task-level autonomy [15]. These systems provide valuable insights into bimanual control, perception, and learning-based manipulation.

However, these platforms rely on high-cost hardware, extensive sensing, and complex software architectures, which limits their practicality for real-world assistive deployment. Their size, cost, and maintenance requirements confine their use primarily to laboratory settings, reducing accessibility and reproducibility.

2.3.4 Summary of Strengths and Limitations

Existing assistive robotic systems exhibit important strengths, including reliable manipulation, autonomous navigation, and advanced bimanual control. However, these capabilities are rarely combined within a single platform that is affordable, modular, and suitable for everyday assistive use. Single-arm systems prioritise simplicity and reliability at the cost of functional completeness, while research-oriented bimanual platforms demonstrate rich manipulation capabilities but lack accessibility and practicality for real-world deployment.

2.4 ALOHA and Mobile ALOHA

ALOHA is a low-cost, open-source hardware system for bimanual manipulation and learning, first developed by Tony Z. Zhao et al. at Stanford University. It has played a key role in the development and validation of ACT [5], representing a significant advancement in imitation learning for robotic systems.

The ALOHA platform is relatively lightweight, requires modest training resources, and enables efficient policy learning, with models typically on the order of 80 million parameters. These characteristics make it a practical and accessible system compared to many recent large-scale robotic learning approaches.

Despite being considered a low-cost system within research contexts, mobile ALOHA configurations still cost on the order of \$20,000–\$30,000, which remains prohibitively expensive for many real-world assistive applications. This highlights a gap between research-grade systems and truly accessible robotic platforms.

As such, ALOHA serves as a strong and appropriate baseline for this project. The objective is to achieve comparable manipulation performance on similar tasks while maintaining a lower overall system cost. Demonstrating similar capabilities at a reduced price point would indicate meaningful progress toward more accessible mobile bimanual robotic systems, directly addressing this gap in the current state of the art.

2.4.1 ACT

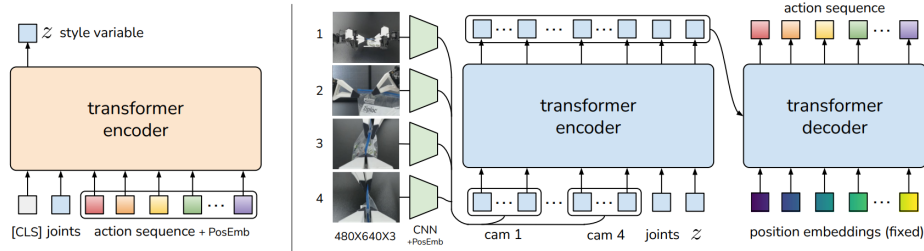


Figure 2.2: ACT policy architecture, where visual observations and robot state are encoded to predict a chunk of future actions, enabling short-horizon planning and continual replanning during execution.

ACT is an imitation learning architecture introduced by Zhao et al. to address challenges associated with long-horizon dexterous manipulation tasks [5]. Conventional behavioural cloning policies predict a single action at each timestep, making them susceptible to compounding errors during execution. Small deviations from demonstrated trajectories can accumulate over time, causing the robot to encounter states not represented within the training data and ultimately resulting in task failure [16]. These issues become particularly pronounced in multi-stage manipulation tasks, where successful completion depends upon the accurate execution of many sequential actions.

To mitigate these limitations, ACT predicts a sequence of future actions, referred to as an *action chunk*, in place of a single control command. Given observations at time t , the policy predicts

$$a_{t:t+k} = [a_t, a_{t+1}, \dots, a_{t+k}],$$

where k denotes the chunk length. Each prediction therefore represents a short-horizon plan describing the intended future behaviour of the robot.

Figure 2.3 illustrates the action chunking paradigm employed by ACT. Consecutive chunks overlap in time, allowing the policy to continually refine future actions as new observations become available.

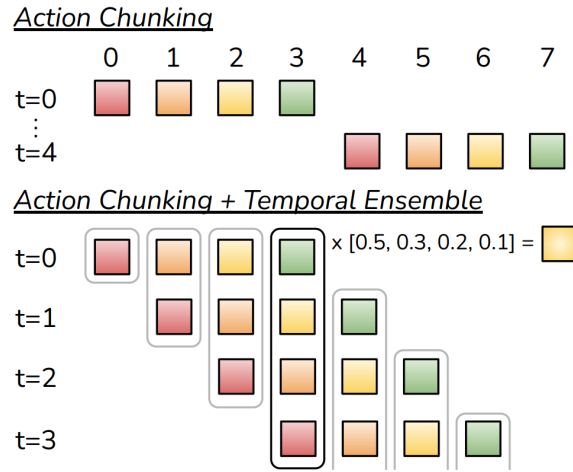


Figure 2.3: ACT action chunking improves long-horizon manipulation by predicting overlapping sequences of future actions as opposed to isolated single-step commands. Successive chunks allow the policy to continually refine its plan as new observations are received.

The overlap between predicted chunks enables a second mechanism referred to as *temporal ensembling*. Since multiple chunks may contain predictions corresponding to the same execution timestep, ACT combines these predictions using exponentially weighted averaging:

$$\hat{a}_t = \frac{\sum_{i=1}^N w_i a_t^{(i)}}{\sum_{i=1}^N w_i},$$

where $a_t^{(i)}$ denotes the prediction originating from the i -th chunk and w_i is an exponentially decaying weighting term [5]. Temporal ensembling reduces the influence of noisy individual predictions and encourages smoother execution trajectories.

ACT employs a transformer-based encoder–decoder architecture conditioned on multimodal observations consisting of visual inputs and robot state information. During training, a conditional variational autoencoder (CVAE) is incorporated to capture variability within human demonstra-

tions. The latent variable encourages the policy to represent multiple valid strategies for completing the same task, while inference is performed using a fixed prior to enable deterministic execution [5].

The ACT architecture demonstrated strong performance across a range of bimanual manipulation tasks on the ALOHA platform, including object transfer, container manipulation, and tool-use behaviours [5]. Its ability to learn from relatively small demonstration datasets, together with its compatibility with low-cost robotic hardware, has resulted in its widespread adoption within the LeRobot ecosystem.

ACT was selected as the baseline policy architecture for this work for three reasons. Firstly, its chunked prediction paradigm is well suited to the long-horizon assistive tasks considered throughout this dissertation. Secondly, its use of a CVAE enables variability in demonstrations to be modelled without requiring prohibitively large datasets. Finally, its demonstrated success on bimanual manipulation tasks provides a strong foundation upon which modifications to the coordination structure of the policy can be investigated.

2.5 LeRobot Platform

LeRobot was selected as the software framework for this project because it provides infrastructure for dataset recording, policy training, and deployment within a common imitation-learning pipeline. This reduced the need to develop a complete learning system from scratch and made the resulting platform more reproducible. In this project, LeRobot therefore served not only as a control framework, but also as the link between teleoperation, demonstration collection, policy training, and autonomous execution.

2.5.1 XLeRobot

The XLeRobot platform was selected as the base hardware system due to its open-source nature, modular design, and suitability for low-cost experimentation. This section outlines the key characteristics of the platform and evaluates its appropriateness for a mobile bimanual assistive robotic application.

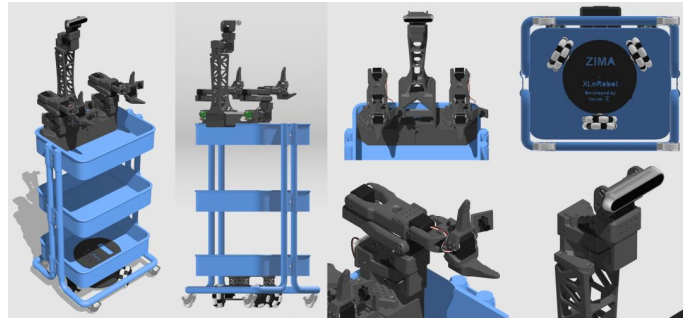


Figure 2.4: XLeRobot provides an affordable open-source foundation for mobile manipulation, but requires extension to support the accessible bimanual assistive behaviours targeted in this work.

2.5.2 Overview of the XLeRobot System

XLeRobot is an open-source robotic platform designed to facilitate research and experimentation in mobile manipulation [6]. The platform typically consists of a mobile base and one or more robotic arms, with an emphasis on accessibility, customisability, and community-driven development.

2.5.3 Capabilities and Limitations

The XLeRobot platform offers sufficient capability for basic manipulation and navigation tasks, including arm control, mobile base movement, and sensor integration [6]. However, as a low-cost system, it also presents limitations in terms of payload capacity, precision, and sensing fidelity [6].

These limitations require careful task selection and control strategy design. As opposed to relying on high-precision hardware, this project leverages learning-based methods and task-level autonomy to achieve functional performance within the platform’s constraints.

2.6 Human–Robot Interaction Considerations for Assistive Use

For assistive robotics, safety, usability, and user agency are critical. Physical interaction requires conservative limits on joint speed, torque, and applied forces to minimise injury risk [17]. Interfaces must also be intuitive and accessible, reducing cognitive load and allowing users to request assistance without requiring specialist robotics knowledge [7].

Prior research in HRI highlights the importance of maintaining user involvement and task

relevance to sustain engagement beyond initial novelty [7]. These principles inform the interaction and control strategies used in this project. Instead of treating the robot as a fully autonomous replacement for the user, the system is designed to provide task-level assistance while preserving accessible methods for user input, supervision, and control.

2.7 Research Gap and Project Motivation

While assistive robotics has demonstrated the potential to support individuals with motor impairments, existing systems are often clinically oriented, or reliant on costly hardware platforms. Single-arm assistive systems provide useful reaching and grasping support, but remain limited for tasks that require stabilisation or coordinated two-handed interaction. Research-oriented bimanual platforms demonstrate more advanced manipulation capabilities, but are often expensive, complex, and impractical for wider deployment.

Existing approaches therefore leave a gap for a low-cost platform that combines mobile bimanual manipulation, learning from demonstration, and accessible user interaction within a single system. This gap is important for assistive contexts, where affordability, ease of use, adaptability, and reproducibility are as important as absolute manipulation performance.

This project addresses this gap by investigating whether a low-cost mobile bimanual robotic assistant can support representative assistive manipulation tasks relevant to users with motor impairments. By focusing on affordability, functional assistance, and accessible interaction, this work aims to contribute toward assistive robotic systems that are technically feasible, reproducible, and more practical for use beyond specialised robotics laboratories.

3

System Objectives

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3.1 System Objectives

The design of the RIGBY platform is guided by four overarching design goals that inform all hardware and software decisions throughout this project. These goals provide a framework for evaluating design choices and measuring the success of the final system.

3.1.1 Low Cost

A primary objective of this project is to determine whether a capable mobile manipulation platform for imitation learning and assistive robotics can be developed at substantially lower cost than existing research systems. To improve accessibility and reproducibility, the target hardware cost for RIGBY was constrained to less than £2,000, excluding development equipment and fabrication tools. This objective informed hardware selection throughout the project, favouring commercially

available components, consumer-grade electronics, and 3D-printed mechanical structures, consistent with previous low-cost open-source robotics platforms that use printable parts and off-the-shelf components to improve reproducibility and ease of repair [18].

3.1.2 Capability

Capability refers to the robot's ability to perform meaningful manipulation tasks and interact effectively with its environment. The system should support mobile navigation, bimanual manipulation, teleoperation, vision-based perception, and autonomous task execution through imitation learning. Design decisions that improve task performance, manipulation dexterity, or autonomy contribute towards this objective.

3.1.3 Accessibility

Accessibility refers to reducing barriers to system operation by making the robot intuitive to use, straightforward to interact with, and reproducible by other researchers. The system should support multiple interaction modalities and minimise the technical expertise required for operation and deployment. Design decisions that improve usability, simplify setup, or support a wider range of users contribute towards this objective.

3.1.4 Robustness

Robustness refers to ensuring safe, predictable, and reliable operation in the presence of uncertainty, disturbances, or hardware failures. The platform should maintain stable operation during teleoperation and autonomous execution while remaining easy to diagnose and repair. Design decisions that improve reliability, fault tolerance, maintainability, or operational safety contribute towards this objective.

The hardware and software objectives presented in the following sections are derived from one or more of these design goals.

3.2 Hardware Objectives

The primary hardware objectives of this project are as follows:

-
- Design and integrate a mobile robotic platform with two manipulators capable of coordinated bimanual manipulation and mobile manipulation tasks.
 - Develop a low-cost teleoperation interface that enables intuitive control of both manipulators and the mobile base.
 - Integrate wrist-mounted and external cameras to provide visual information for teleoperation, perception, and autonomous operation.
 - Ensure the hardware is modular and reproducible, supporting maintenance, upgrades, and replication of the platform.
 - Select components that provide a suitable balance of cost, payload capacity, reliability, and overall system performance.

3.3 Software Objectives

- Enable teleoperation of the mobile manipulation platform for demonstration collection.
- Support the training and deployment of imitation learning policies.
- Facilitate coordinated manipulation behaviours involving both manipulators and the mobile base.
- Provide simulation support for development and policy validation.
- Develop an accessible interface for system monitoring, demonstration recording, and policy execution.
- Support multiple user interaction modalities, including graphical, textual, and voice-based inputs.
- Enable task-level interaction through natural user requests while maintaining separation from low-level robot control.
- Maintain a modular software architecture that supports future extension and integration.

4

Analysis and System Design

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This chapter translates the project objectives into the main design decisions that shaped RIGBY. Instead of describing implementation details directly, it first explains the reasoning behind the chosen system architecture, including the decision to build on an open-source platform, use imitation learning, collect demonstrations through teleoperation, and adopt an asymmetric bimanual configuration. These decisions provide the foundation for the hardware and software implementation described in the following chapters.

4.1 Analysis and System Design

4.1.1 Design Philosophy and Scope of Contribution

The initial objective of this project was to evaluate whether a low-cost mobile bimanual robot could support assistive manipulation tasks relevant to people living with motor impairments. Early investigation considered the development of a completely bespoke robotic platform. However, this approach was deemed impractical within the time and resource constraints of the project.

Instead, an existing open-source platform, XLeRobot, was adopted as the foundation for development. This decision aligned with the objectives of affordability, reproducibility, and rapid

prototyping. Nevertheless, the final system differs substantially from the original platform. Modifications included the introduction of asymmetric manipulators, a whole-body teleoperation interface, a multimodal interaction framework, a simulation environment, and a novel imitation learning architecture.

Consequently, the project evolved from adapting an existing robot into the development of an integrated assistive robotics research platform focused on accessibility, extensibility, and demonstration-based learning.

4.1.2 Key System-Level Design Decisions

Several important design decisions shaped the final system architecture. In each case, alternative approaches were considered and evaluated against the overarching project objectives of affordability, accessibility, and functional capability.

Open-source platform versus bespoke design

A fully custom robot would maximise design freedom but would significantly increase development effort and reduce reproducibility. Building upon XLeRobot enabled resources to be focused on the aspects of greatest research value, including teleoperation, interaction design, and learning-based control.

This approach is consistent with broader trends within robotics research, where open-source platforms have been shown to accelerate development, improve reproducibility, and reduce barriers to entry [19]. In contrast, bespoke robotic systems provide greater design flexibility but often require substantial engineering effort that may limit accessibility and replication [20]. The resulting trade-off favoured extensibility and reproducibility over complete mechanical customisation.

Imitation learning versus reinforcement learning

Several learning paradigms were considered for enabling autonomous behaviour acquisition. Reinforcement learning has demonstrated impressive performance on a range of manipulation benchmarks by enabling agents to learn through trial-and-error interaction with their environment [21]. However, such approaches frequently require large numbers of interactions, making deployment directly on physical robots time-consuming and potentially detrimental to hardware longevity [22].

Imitation learning was therefore selected as the primary learning paradigm. By learning directly from demonstrations, imitation learning substantially reduces the amount of physical interaction required while maintaining human oversight throughout the teaching process. This characteristic aligns particularly well with assistive applications, where demonstrations naturally encode user preferences and task expectations.

Furthermore, the use of lower-cost actuators motivated a preference for demonstration-based learning. Extensive exploration associated with reinforcement learning may accelerate mechanical wear, increase maintenance requirements, and introduce safety concerns when deployed on physical systems. Although reinforcement learning remains a promising direction for future work, imitation learning provided a more practical and reliable approach within the constraints of this project.

Teleoperation versus kinesthetic teaching

Several approaches for collecting demonstrations were considered. Kinesthetic teaching was unsuitable due to the limited back-drivability of the selected actuators, while autonomous data generation conflicted with the objective of maintaining user involvement.

Leader-follower teleoperation was therefore selected, allowing demonstrations to be collected intuitively while preserving direct human control over task execution.

Leader-follower teleoperation has been widely adopted for collecting high-quality demonstrations for imitation learning, particularly in mobile manipulation settings [19]. By contrast, kinesthetic teaching is often more suitable for compliant and back-drivable manipulators [23], making it less appropriate for systems employing lower-cost actuators.

Symmetric versus asymmetric manipulation

Initial designs employed two identical manipulators. However, preliminary task analysis revealed that many target activities involved complementary arm roles, where one manipulator performed the primary action while the second provided support.

An asymmetric configuration was therefore adopted, allocating additional dexterity to the dominant arm while reducing overall hardware complexity and cost.

This decision reflects observations from studies of human bimanual manipulation, which suggest that complementary instead of identical roles frequently emerge during coordinated tasks [24].

However, symmetric bimanual platforms remain common within robotics due to their versatility and simplified control formulations [25]. The chosen design therefore prioritised task-specific efficiency over complete interchangeability between manipulators.

Conventional interfaces versus multimodal interaction

Traditional robotics systems often rely on command-line interfaces, creating barriers for non-technical users. To improve accessibility, the system was designed to support graphical, textual, and voice-based interaction modalities.

Natural language was incorporated as a mechanism for task selection as opposed to direct robot control, ensuring that robot behaviour remained constrained to validated policies.

Multimodal interfaces have been shown to improve accessibility and usability within assistive robotics applications by accommodating diverse user needs and capabilities [26]. Nevertheless, unrestricted natural language control may introduce ambiguity and safety concerns, motivating the use of constrained task-level interaction paradigms [27]. The resulting design balances usability with the predictability required for assistive applications.

5

Hardware Design and Implementation

Contents

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This chapter describes the physical implementation of the RIGBY platform and its teleoperation system. The hardware design was guided by the need to balance affordability, reproducibility, manipulation capability, and accessibility. The chapter first presents the robot platform, including its manipulators, mobile base, sensing hardware, interaction hardware, power distribution, and communication architecture. It then describes the teleoperator used to collect demonstrations for imitation learning.

5.1 Robot Design

The robot platform was designed as a low-cost mobile bimanual system capable of supporting both teleoperation and autonomous policy execution. Its physical structure combines a mobile base, two

manipulators, onboard sensing, local computation, and a user-facing interface within a compact platform suitable for assistive manipulation experiments. Figure 5.1 provides an overview of the final system configuration and key specifications.

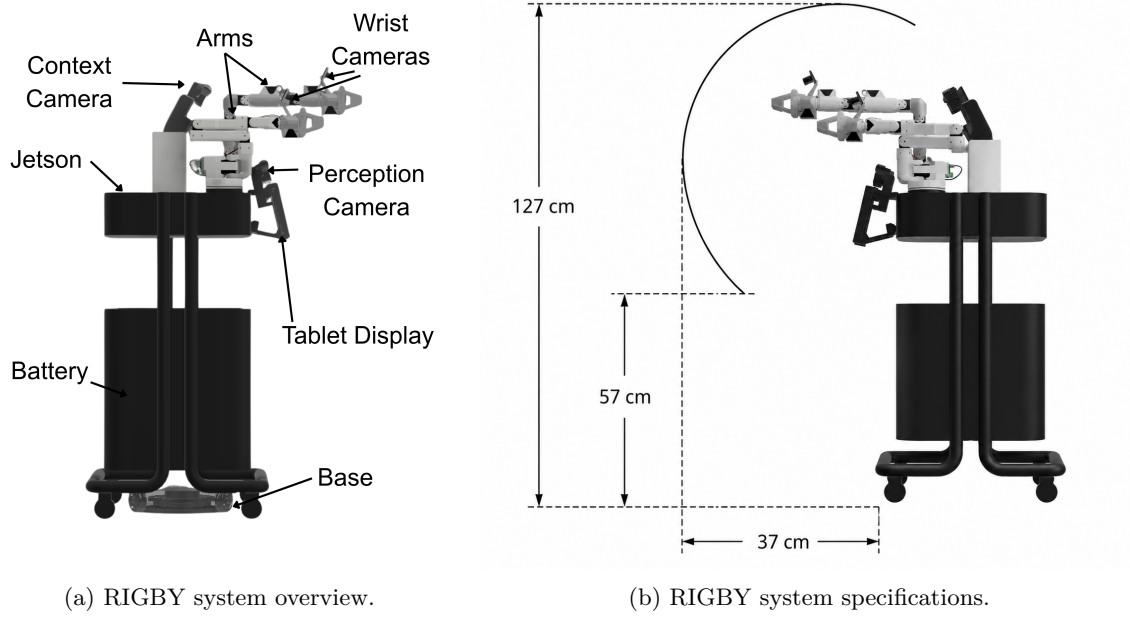


Figure 5.1: RIGBY integrates a low-cost mobile base, asymmetric bimanual arms, onboard sensing, and user-facing interaction hardware into an accessible mobile manipulation platform.

Table 5.1: RIGBY system specifications.

Parameter	Value
DoFs	12 (arms) + 2 (base) + 2 (grippers)
Weight	75 kg
Size	45 W $\times$ 72 L $\times$ 127 H cm
Payload	240 g (per arm), 25 kg (base)
Arm Repeatability	2–4 mm
Battery Life	2 hours continuous moderate use (288 Wh)
Control Frequency	Up to 50Hz

5.1.1 Arms

Placement and Configuration

The initial design adopted the XLeRobot configuration proposed by Vector Wangel, comprising two symmetric 5-DOF LeRobot SO101 manipulators mounted at the rear of the trolley. Early teleoperation experiments revealed limitations in both the available workspace and the ability to manipulate larger household objects. Consequently, both manipulators were repositioned to the front of the platform to improve the robot’s capability.

In parallel, the original symmetric manipulator arrangement was reconsidered, leading to the development of the aforementioned asymmetric bimanual configuration used in RIGBY. The resulting platform employs a dominant left arm and a secondary right arm, shown in

Table 5.2: Manipulator configurations employed in the RIGBY platform.

Manipulator	DOF	Motors	Function
Dominant arm	7	8	Dexterity and orientation flexibility
Support arm	5	6	Support and repositioning

The asymmetric design was motivated by the dominant-support interaction patterns observed during the target assistive tasks, where one manipulator frequently performed the task-defining action while the other provided stabilisation or environmental support. Concentrating the additional dexterity on the dominant manipulator therefore reduced hardware complexity and cost while maintaining effective bimanual functionality. To maintain compatibility with the LeRobot framework, all manipulator joints utilised FEETECH STS3215 [28] serial bus servos.

The geometry of the dominant 7 DOF arm was inspired by an open-source concept proposed by Norma Core, however the final mechanical design was developed specifically for this project. Particular attention was given to the relationship between arm length, joint placement, and mass distribution.

Early iterations employing longer links and heavier distal components produced excessive joint loading and reduced teleoperation usability. The final design therefore balanced reach, mass, and structural efficiency to improve manoeuvrability during demonstration collection while maintaining sufficient workspace for assistive manipulation tasks.

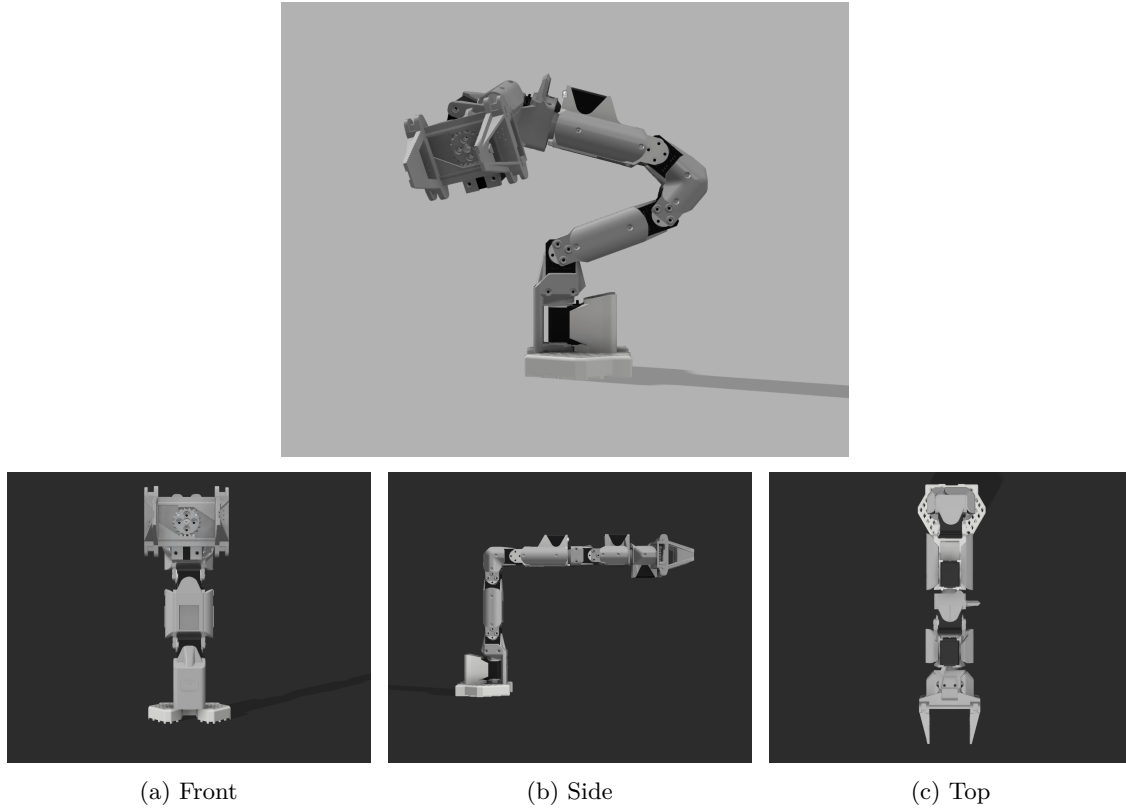


Figure 5.2: Multi-view assembly of the left arm shows how the added degrees of freedom increase dominant-arm dexterity while maintaining a compact low-cost mechanical design.

End-Effectors

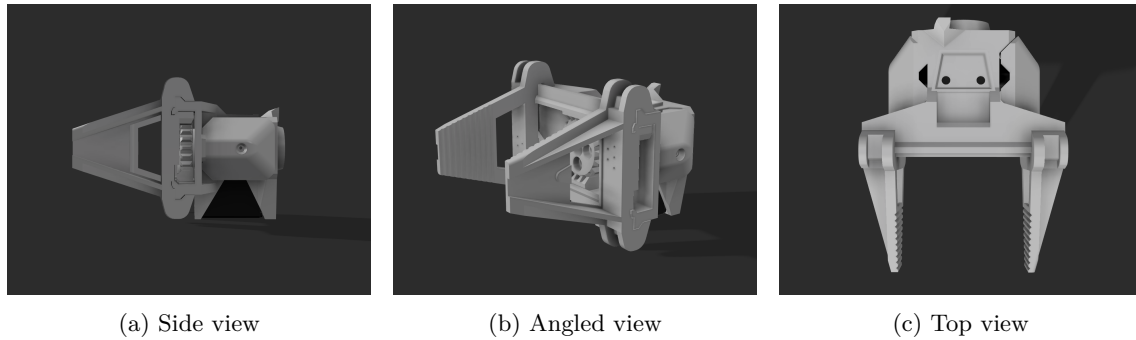


Figure 5.3: The redesigned parallel gripper reduces wrist-camera occlusion through visibility cut-outs while preserving grasp surfaces for household object manipulation.

The original LeRobot SO101 grippers were replaced with parallel-jaw grippers based on the Norma Core PGripper [29] design to improve grasp reliability across a wider range of household objects. The gripper design was subsequently refined to better support vision-based imitation learning, as sections of the original finger geometry frequently occluded the wrist-mounted camera view during grasping and fine manipulation tasks. To mitigate this issue, the gripper fingers were redesigned

with integrated visibility cut-outs that reduced visual occlusion while maintaining adequate grasp stability (Figure 5.3). This modification improved the quality of recorded demonstrations by providing the wrist-mounted cameras with a clearer view of object interactions during manipulation.

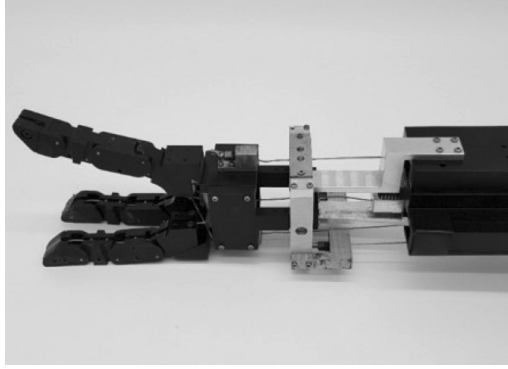


Figure 5.4: Example of an adaptive underactuated hand considered during end-effector selection. Although such designs can improve grasp conformity around varied objects, their enclosing finger structure was less suitable for RIGBY because it risked increasing wrist-camera occlusion during vision-based manipulation [30].

An adaptive underactuated hand, such as the open-source 3D-printed design proposed by Ma et al. [30], was also considered. Although such designs can improve grasp conformity, their larger enclosing finger structure was likely to increase wrist-camera occlusion. A simpler parallel-jaw gripper with visibility cut-outs was therefore preferred to preserve clear visual observations for imitation learning.

5.1.2 Mobile Body

Following the original XLeRobot design proposed by Vector Wangel, the robotic platform utilises an IKEA RÅSKOG utility cart mounted on a LeKiwi omnidirectional mobile base [31], [32]. The cart provides a low-cost and easily reproducible mounting structure for the manipulators, sensors, computing hardware, and power system while maintaining a suitable working height for common household tasks.

The LeKiwi base enables omnidirectional motion, allowing the robot to translate in any planar direction while simultaneously rotating about its vertical axis. This significantly extends the effective workspace of the manipulators and supports whole-body behaviours requiring coordinated navigation and manipulation. Such capabilities are particularly beneficial during teleoperation and imitation learning, where frequent positional adjustments are often required during demonstration collection [33].

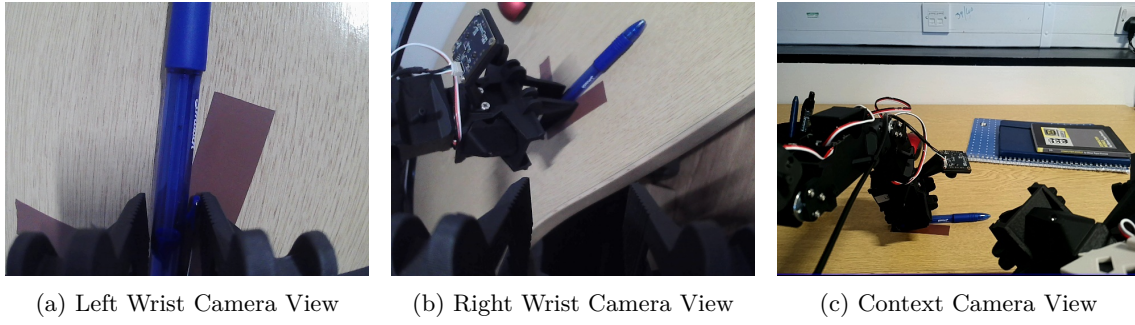


Figure 5.5: The onboard wrist and context camera views provide complementary local and global observations required for vision-based demonstration learning.

5.1.3 Sensors and User Interaction Hardware

The sensing and interaction architecture was designed to support imitation learning, autonomous execution, and accessible human-robot interaction. Building upon the XLeRobot platform, the perception system was extended to incorporate additional sensing and interaction capabilities developed specifically for this project. The resulting configuration comprises four cameras, a USB microphone array, and a front-mounted 7-inch touchscreen display.

Each manipulator is equipped with an Innomaker 1080P wrist-mounted camera positioned adjacent to the gripper. These cameras provide close-range observations of object interactions during demonstration collection and policy execution. Two OAK-D RGB-D cameras were additionally incorporated into the platform. The first serves as a context camera, providing a wide-angle view of the surrounding workspace to complement the wrist-camera observations during imitation learning. The second functions as an interaction camera, supplying depth-aware observations of the robot’s immediate operating environment and supporting future perception-driven behaviours. To facilitate multimodal interaction, RIGBY incorporates a USB microphone array and a front-mounted 7-inch touchscreen display. The microphone array enables future voice-based interaction, while the touchscreen provides the primary user interface for presenting system status information, task controls, and feedback during teleoperation and autonomous operation.

5.1.4 Electronics and Power Distribution

The platform was designed for fully untethered operation using an Anker SOLIX C300 portable power station. The C300 provides both AC and high-power USB-C Power Delivery outputs with differing power capabilities, requiring subsystem loads to be distributed accordingly.

Power allocation decisions were informed using conservative estimates derived from manufacturer specifications [28], [34], [35]. As summarised in Table 5.3, the highest-demand manipulator was supplied directly from an AC output, while lower-power computational components shared the remaining AC supply. The mobile base and Jetson Orin Nano were powered using the high-power USB-C outputs. This load-aware distribution strategy enabled reliable untethered operation while maintaining the portability and modularity objectives of the platform.

Table 5.3: Estimated worst-case subsystem loads used to inform power allocation decisions. Servo estimates are derived from FEETECH STS3215 manufacturer specifications.

Subsystem	Actuator	Qty.	P_{peak}	SOLIX output
Left follower arm	STS3215	8	259.2 W	AC output 1
Right follower arm	STS3215	6	194.4 W	AC output 2
Mobile base	STS3215	3	97.2 W	USB-C PD
Jetson Orin Nano	Compute	1	≈ 25 W	USB-C PD
Powered USB hub	Hub	1	Low auxiliary load	AC output 2

Communication Architecture

An Nvidia Jetson Orin Nano was used as the central compute and communication node for the platform. To simplify wiring and improve maintainability, peripheral devices were connected through a powered USB hub wherever possible. This provided a single aggregation point for serial devices, cameras, the pedal interface, and the touchscreen display, while allowing each subsystem to remain independently replaceable and debuggable.

The follower arms, leader arms, and mobile base communicated using independent USB serial links. This separation avoided coupling between actuation subsystems and allowed individual devices to be identified during calibration and troubleshooting. The pedal interface used a standard USB HID joystick connection, while the wrist and context cameras used USB video interfaces for demonstration recording and policy inference. The 7-inch touchscreen display provided the local user interface for task selection, recording, diagnostics, and system feedback.

Table 5.4 summarises the principal communication links.

5.2 Teleoperator Design

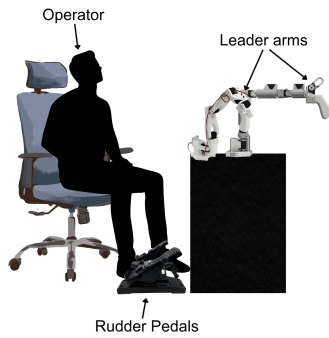
The teleoperator was developed to provide an intuitive method for collecting demonstrations for imitation learning. The design mirrors the physical configuration of RIGBY so that operator

Table 5.4: Summary of communication interfaces used throughout the platform.

Subsystem	Interface Type
Left follower arm	USB serial
Right follower arm	USB serial
Mobile base	USB serial
Left leader arm	USB serial
Right leader arm	USB serial
Pedal interface	USB HID
Wrist cameras	USB video
Context camera	USB/OAK interface
7-inch touchscreen display	HDMI/USB touch
Powered USB hub	USB uplink

movements correspond closely to the robot’s action space. This allows demonstrations to be recorded as joint and base commands that can be replayed or used directly within the LeRobot data collection pipeline.

Figure 5.6 provides an overview of the teleoperator system.



(a) Diagram of teleoperation system



(b) Physical teleoperation system

Figure 5.6: The teleoperator platform mirrors the robot’s whole-body control structure, allowing demonstrations to be collected through intuitive leader-follower operation.

5.2.1 Leader Arms

The teleoperator uses the same asymmetric bimanual structure as the robot, consisting of a dominant left leader arm and a secondary right leader arm. Matching the kinematic structure of the robot in the teleoperator reduces the mapping complexity between the operator input and robot motion during demonstration collection.

This design was selected to support natural bimanual teleoperation while preserving the asymmetry introduced in the robot hardware. Since the left arm is intended to perform more dexterous

manipulation and the right arm is primarily used for stabilisation or support, the teleoperator preserves the same division of roles. This helps the operator demonstrate tasks in a way that is consistent with the physical capabilities of the robot and with the assumptions used in the learning architecture.

5.2.2 End-Effectors

Unlike the robot arms, which use modified parallel jaw grippers for improved object retention and camera visibility, the teleoperator arms use the standard trigger mechanism from the SO101 leader arm design. These triggers act as the operator input for opening and closing the corresponding robot grippers.

This separation between leader and follower end-effectors was chosen because the teleoperator does not need to physically grasp objects. Instead, it only needs to provide a reliable and comfortable input signal representing the operator’s intended gripper state. The SO101 trigger mechanism provides a compact and intuitive control input, allowing the user to command grasping actions without adding unnecessary mechanical complexity to the teleoperator.

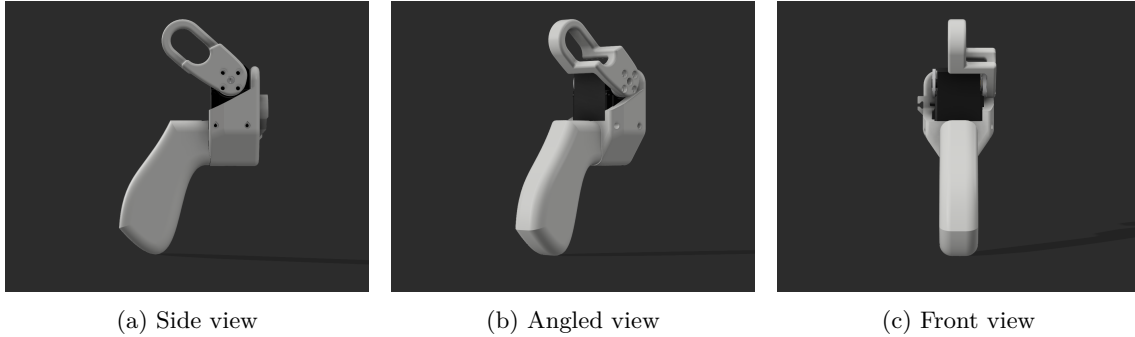


Figure 5.7: The SO101 trigger mechanism converts natural finger input into gripper commands, improving the usability of demonstration collection.

An alternative teleoperator end-effector design was also designed and considered, based on a parallel-jaw mechanism with integrated finger slots, as shown in Figure 5.8. This design was intended to make gripper control more physically intuitive by allowing the operator to actuate the mechanism using a pinching motion similar to the robot gripper. However, preliminary testing showed that this arrangement required excessive finger torque and was less comfortable for repeated demonstration collection. The standard SO101 trigger mechanism was therefore retained, as it provided a lighter, more reliable, and more ergonomic input method for commanding gripper opening and closing.

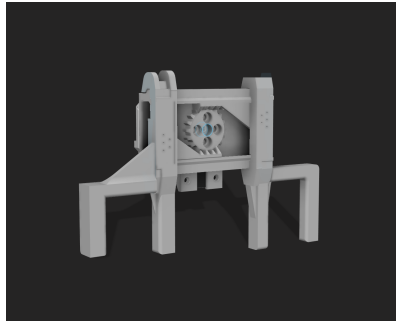


Figure 5.8: Alternative teleoperator end-effector concept using a parallel-jaw mechanism with finger slots. Although intended to provide a more intuitive pinching-style input, the design required excessive finger torque and was therefore rejected in favour of the standard SO101 trigger mechanism.

5.2.3 Base and Pedals

In addition to arm motion, the teleoperator provides direct control over the mobile base. Base movement is controlled using rudder pedals, shown in Figure 5.9, which allow the operator to command translational and rotational motion while keeping both hands available for bimanual arm control.

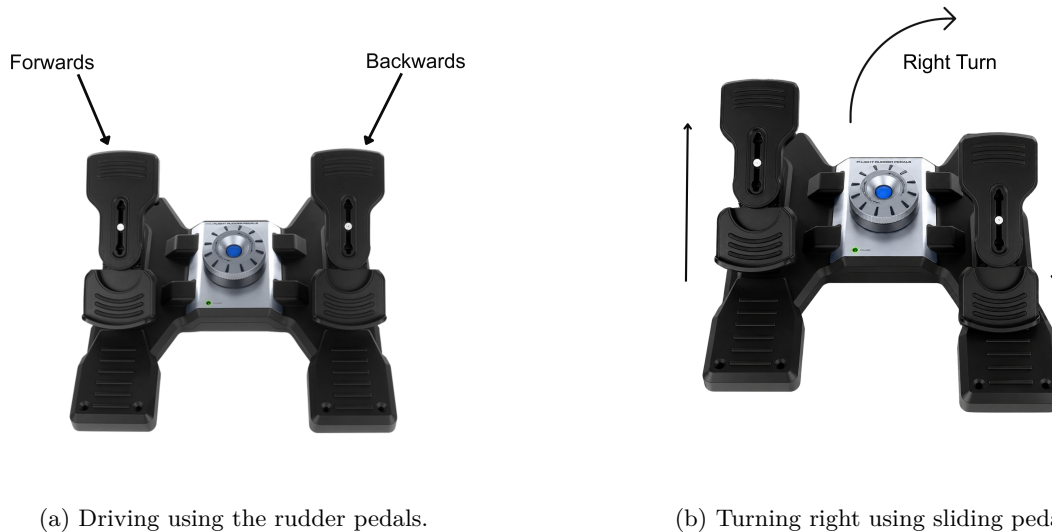


Figure 5.9: The rudder pedal interface separates mobile-base control from arm control, enabling simultaneous whole-body teleoperation during demonstrations.

This separation of control modalities was important because whole-body teleoperation requires simultaneous coordination of the arms and mobile base and consistent with recent low-cost whole-body teleoperation systems [36]. Assigning base motion to the feet allows the operator to reposition the robot while continuing to manipulate objects with both leader arms. This is particularly useful during long-horizon tasks where the robot must approach a workspace, adjust its position, and

continue manipulation without interrupting arm control.

The pedal interface therefore improves demonstration collection by enabling coordinated mobile manipulation. It also reflects the role of the mobile base in the robot system, where small positional adjustments can significantly improve reachability and task success.

5.2.4 Control Electronics

The teleoperator uses a modular USB-based communication architecture similar to that of RIGBY. Dedicated USB connections are used for the leader arms and pedal interface, simplifying device identification and troubleshooting during demonstration collection.

Unlike the robot platform, the teleoperator is powered directly from mains electricity instead of an onboard battery. Communication is managed through a powered USB hub, which connects all input devices to the host computer. This architecture provides a simple and reliable platform for teleoperation and imitation learning data collection.

6

Software Design and Implementation

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This chapter presents the software components that enable RIGBY to function as an end-to-end imitation learning platform. The implementation connects robot control, teleoperation, demonstration recording, policy training, autonomous execution, simulation, and user interaction within a single workflow. Emphasis is placed on making the system accessible to operate while preserving the modularity required for future research and extension.

6.1 System Design

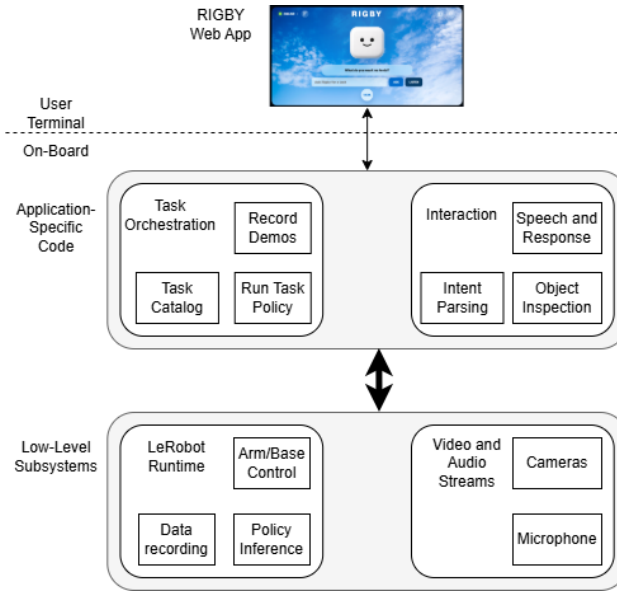


Figure 6.1: System-level architecture of the RIGBY software stack. The user-facing web application communicates with application-specific services for task orchestration, demonstration recording, policy execution, intent parsing, speech interaction, and object inspection. These services are built on top of the LeRobot runtime and sensor interfaces, which provide data recording, policy inference, robot command execution, and access to camera and audio streams.

The RIGBY software system was designed as a collection of interconnected components supporting teleoperation, demonstration recording, policy training, autonomous execution, and user interaction.

At its core, the Jetson Orin Nano coordinates communication between the robot hardware, perception system, learning framework, and touchscreen interface. Demonstrations collected through the teleoperation system are processed using the LeRobot framework to train imitation learning policies, which can subsequently be selected and executed through the RIGBY user interface. Additional services, including voice interaction and local language model inference, provide a natural language layer for task selection while maintaining separation between user intent and low-level

robot control.

6.2 Manipulation Implementation

6.2.1 Learning

The learning and teleoperation systems in this project were designed with two objectives. The first was to allow the user to efficiently collect high-quality demonstration data for imitation learning. The second was to ensure the overall system remained accessible and reproducible for future research and deployment. To this end, the software was built around the LeRobot ecosystem [37], while extending it with custom interfaces, policy architectures, and simulation support specifically designed for RIGBY’s asymmetric bimanual manipulation.

Teleoperation and Data Collection

The teleoperation and data collection pipeline is based on the LeRobot recording framework, adapted for the custom RIGBY hardware platform. This included modifications to support the asymmetric bimanual arm configuration, custom motor interfaces, mobile base control, and the multimodal camera setup used throughout the system. The resulting pipeline enabled synchronous collection of joint states, base commands, visual observations, and operator demonstrations during teleoperation.

6.2.2 Imitation Learning Policies

Following demonstration collection, policies were trained using imitation learning to enable the robot to reproduce behaviours directly from expert demonstrations. This work adopts the ACT framework as the baseline policy architecture due to its demonstrated effectiveness in learning long-horizon manipulation tasks from relatively small demonstration datasets. ACT predicts sequences of future actions conditioned on multimodal observations, improving robustness to compounding errors during execution.

However, the asymmetric hardware configuration employed by RIGBY motivated investigation into whether additional inductive biases could improve learning efficiency for dominant-support manipulation tasks. Consequently, this work explores an extension of ACT, referred to as DOM-ACT,

which explicitly models the relationship between the dominant manipulator, support manipulator, and mobile base within a unified policy architecture.

6.2.3 DOM-ACT

Motivation

Many mobile bimanual manipulation tasks exhibit an asymmetric structure in which one manipulator performs the primary task-defining action while the remaining degrees of freedom provide support through stabilisation, repositioning, or navigation. Examples include pen insertion, where one arm performs the grasping and insertion behaviour while the second manipulator repositions the workspace, and ziploc closure, where one arm executes the sealing motion while the other maintains tension. Conventional ACT policies predict all action dimensions simultaneously from a shared latent representation and therefore do not explicitly model these relationships.

DOM-ACT introduces a dominant-support inductive bias into the policy architecture. The dominant manipulator is responsible for determining the primary manipulation trajectory, while the support manipulator and mobile base adapt their behaviour relative to this trajectory. Within this work, the left arm is treated as the dominant manipulator to align with the demonstration strategy employed during data collection. Consequently, DOM-ACT should be interpreted as a task-specific extension of ACT intended for asymmetric manipulation tasks instead of a universally applicable replacement.

Policy Architecture

Let the predicted action chunk be partitioned into left-arm, right-arm, and mobile-base trajectories:

$$a_{t:t+k} = [a_{t:t+k}^L, a_{t:t+k}^R, a_{t:t+k}^B],$$

where a^L , a^R , and a^B denote the actions of the dominant left manipulator, support right manipulator, and mobile base respectively.

Similar to ACT, multimodal observations consisting of visual observations and robot state information are encoded into a shared representation. Given observations o_t , the encoder produces

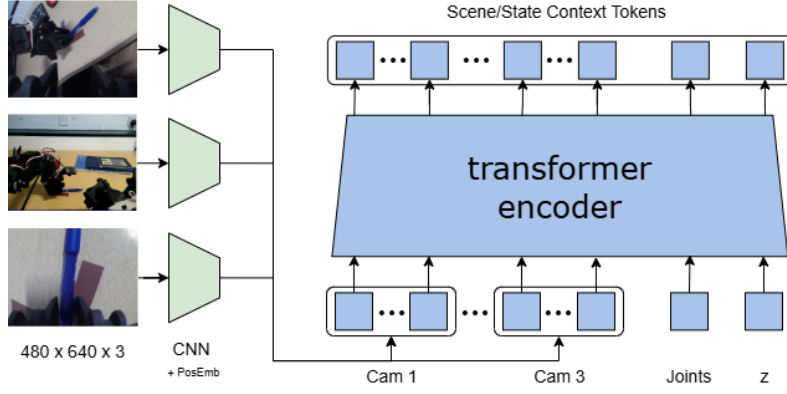


Figure 6.2: DOM-ACT inference introduces a dominant-support structure in which the dominant-arm plan conditions both the support arm and mobile base predictions.

$$h = E(o_t, z),$$

where z denotes the optional variational latent sampled from the ACT-style VAE during training.

The dominant-arm decoder predicts both the left-arm action chunk and an intermediate representation describing the intended manipulation trajectory:

$$(\hat{a}_{t:t+k}^L, h_L) = D_L(h),$$

where $D_L(\cdot)$ denotes the dominant-arm decoder and h_L represents the dominant-arm decoder features.

The mobile base prediction head operates directly on the dominant-arm representation:

$$\hat{a}_{t:t+k}^B = D_B(h_L),$$

encouraging locomotion behaviours to emerge in support of the dominant manipulation objective.

Finally, the support-arm decoder is conditioned on both the shared encoder representation and the dominant-arm features:

$$\hat{a}_{t:t+k}^R = D_R(h, h_L).$$

This dependency structure allows the support manipulator to generate actions relative to the

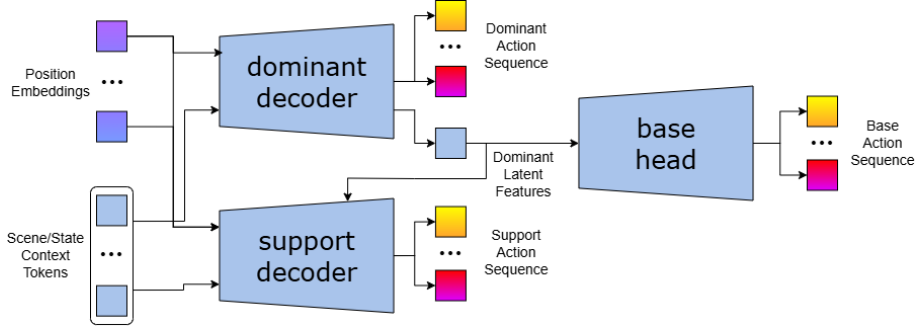


Figure 6.3: DOM-ACT architecture. Position embeddings and scene/state context tokens are processed by separate dominant and support decoders. The dominant decoder predicts the dominant-arm action sequence and produces latent features that condition the support decoder and base head, enabling support-arm and mobile-base actions to be generated relative to the dominant manipulation plan.

intended dominant-arm trajectory as opposed to predicting all action dimensions independently. Consequently, DOM-ACT embeds an explicit dominant-support relationship within the policy architecture, reducing the burden on the network to infer coordination structure solely from demonstration data.

Unlike conventional ACT, which predicts all action dimensions simultaneously from a shared decoder, DOM-ACT introduces a sequential dependency in which the dominant-arm representation influences both support-arm and mobile-base predictions. Nevertheless, the policy retains the chunked execution paradigm of ACT, allowing deployment using the same queue-based inference procedure, as summarised in Algorithm 1.

Algorithm 1 DOM-ACT Inference

1: Initialise FIFO action queue $\mathcal{Q}$ 2: **for** timestep $t = 1, \dots, T$ **do**3: **if** $\mathcal{Q}$ is empty **then**4: Set $z = 0$

5: Encode observations:

$$h = E(o_t, z)$$

6: Predict dominant-arm actions and features:

$$(\hat{a}^L, h_L) = D_L(h)$$

7: Predict mobile-base actions:

$$\hat{a}^B = D_B(h_L)$$

8: Predict support-arm actions:

$$\hat{a}^R = D_R(h, h_L)$$

9: Form action chunk:

$$\hat{a} = [\hat{a}^L, \hat{a}^R, \hat{a}^B]$$

10: Push the next n_{action} actions from $\hat{a}$ into $\mathcal{Q}$ 11: **end if**

12: Execute:

$$a_t = \text{pop}(\mathcal{Q})$$

13: **end for**

Training Objective

DOM-ACT extends the standard ACT training objective by decomposing the action prediction loss across the dominant arm, support arm, and mobile base while additionally encouraging coordinated support behaviour through dedicated regularisation terms. Furthermore, the architecture supports partially supervised demonstrations, enabling left-arm-only demonstrations and fully bimanual demonstrations to be incorporated within a unified training framework.

For each subsystem, a masked L1 loss is computed:

$$\mathcal{L}_X = \frac{\sum_i m_i^X |\hat{a}_i^X - a_i^X|}{\sum_i m_i^X}, \quad X \in \{L, R, B\},$$

where $\hat{a}^X$ and a^X denote the predicted and demonstrated actions respectively, and m^X is a binary supervision mask indicating which action dimensions are present within the current demonstration. This formulation allows demonstrations with differing levels of supervision to contribute meaningful gradients during optimisation.

To encourage smooth support-arm trajectories, an additional temporal regularisation term is introduced:

$$\mathcal{L}_{smooth} = \frac{1}{k-1} \sum_{i=1}^{k-1} \|\hat{a}_{t+i+1}^R - \hat{a}_{t+i}^R\|_1,$$

which penalises abrupt changes in the predicted support-arm motion.

A second regularisation term encourages consistency between the dominant-arm conditioning representation and the support-arm decoder features:

$$\mathcal{L}_{align} = \|h_R - \phi(h_L)\|_2^2,$$

where h_R denotes the support-arm decoder representation and $\phi(\cdot)$ represents the transformation applied to the dominant-arm features prior to conditioning. This term encourages the support manipulator to remain aligned with the dominant-arm trajectory during prediction.

Consistent with ACT, an optional variational objective is retained to improve robustness to multimodal demonstrations:

$$\mathcal{L}_{KL} = D_{KL}(q_\phi(z \mid a_{t:t+k}, o_t) \parallel \mathcal{N}(0, I)).$$

For partially supervised batches, this objective can be selectively disabled to avoid learning latent action distributions from incomplete demonstrations.

The complete DOM-ACT training objective is therefore given by

$$\mathcal{L}_{total} = \lambda_L \mathcal{L}_L + \lambda_R \mathcal{L}_R + \lambda_B \mathcal{L}_B + \lambda_s \mathcal{L}_{smooth} + \lambda_c \mathcal{L}_{align} + \beta \mathcal{L}_{KL},$$

where λ_L , λ_R , and λ_B weight the action prediction losses associated with the dominant arm, support arm, and mobile base respectively, while λ_s , λ_c , and β control the influence of the smoothness, alignment, and variational objectives.

Algorithm 2 summarises the complete optimisation procedure. Relative to ACT, the principal differences arise from the decomposition of action prediction across subsystems, the incorporation of partially supervised demonstrations through masked losses, and the introduction of support-specific regularisation terms.

Algorithm 2 DOM-ACT Training

- 1: Initialise encoder E , dominant decoder D_L , support decoder D_R , and base head D_B
 - 2: **for** each training batch **do**
 - 3: Sample observations o_t , action chunks $a_{t:t+k}$, and supervision masks m
 - 4: **if** VAE enabled for batch **then**
 - 5: Compute:

$$(\mu, \log \sigma^2) = q_\phi(z \mid a_{t:t+k}, o_t)$$
 - 6: Sample:

$$z = \mu + \sigma \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$
 - 7: **else**
 - 8: Set $z = 0$
 - 9: **end if**
 - 10: Encode observations:

$$h = E(o_t, z)$$
 - 11: Predict dominant-arm actions and features:

$$(\hat{a}^L, h_L) = D_L(h)$$
 - 12: Predict mobile-base actions:

$$\hat{a}^B = D_B(h_L)$$
 - 13: Predict support-arm actions:

$$(\hat{a}^R, h_R) = D_R(h, h_L)$$
 - 14: Compute masked action losses:

$$\mathcal{L}_L, \mathcal{L}_R, \mathcal{L}_B$$
 - 15: Compute support regularisation terms:

$$\mathcal{L}_{smooth}, \mathcal{L}_{align}$$
 - 16: Form total objective:

$$\mathcal{L}_{total} = \lambda_L \mathcal{L}_L + \lambda_R \mathcal{L}_R + \lambda_B \mathcal{L}_B + \lambda_s \mathcal{L}_{smooth} + \lambda_c \mathcal{L}_{align} + \beta \mathcal{L}_{KL}$$
 - 17: Update network parameters using AdamW
 - 18: **end for**
-

6.2.4 Simulation and URDF Development

Teleoperation is an effective method for collecting demonstrations, however it requires a dedicated operator and physical access to the robot. This means that the robot must be taken out of normal use during data collection, making it unavailable for other tasks. To add to this, recording sufficiently high-quality demonstrations can require a significant amount of time and familiarity with the teleoperation system. This can cause a problem for users who are unable or unwilling to perform the training process themselves.

To address these limitations and improve accessibility, a complete Unified Robot Description Format (URDF) model of the robotic platform was developed (Figure 6.4). The model captures the asymmetric arm configuration, mobile base, grippers, camera placements, and kinematic constraints of the physical system, enabling deployment within standard robotics simulation environments. This allows tasks to be trained remotely through simulation using imitation learning or reinforcement learning techniques, after which the resulting policy can be transferred to the physical robot. As a result, users can benefit from newly trained capabilities without needing to collect demonstrations locally or dedicate time to learning the teleoperation system.

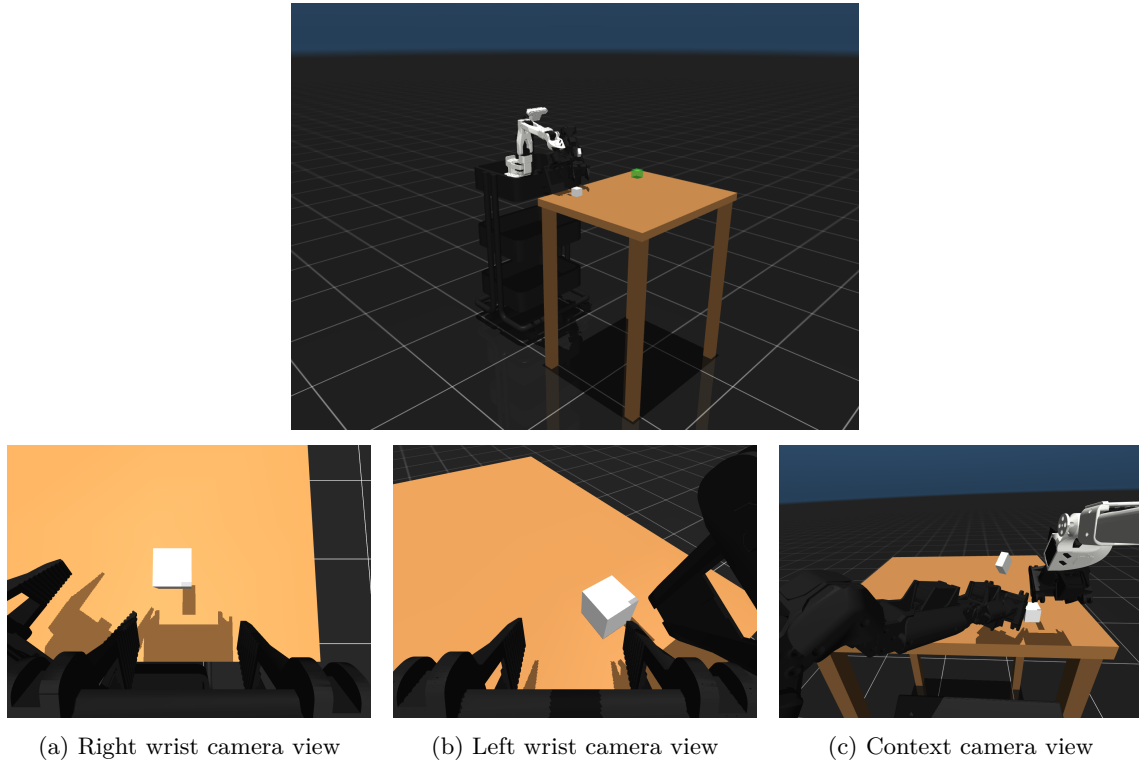


Figure 6.4: The MuJoCo simulation verifies RIGBY’s robot geometry and onboard camera view-points, enabling future simulation-based development before physical deployment.

Beyond supporting the workflow presented in this project, the simulation model extends the utility of the platform for future research. The URDF enables integration with established robotics tools, including ROS2, and simulation environments for reinforcement learning, sim-to-real transfer, motion planning, and autonomous manipulation research without requiring modification of the physical robot. Maintaining compatibility with common robotics tooling therefore supports the broader objective of creating a low-cost, accessible, and reproducible research platform.

6.3 Task Selection and User Interaction

Controlling LeRobot directly from terminals is fragile during demonstrations and reduces accessibility for non-expert users. The objective of the user interface was therefore to provide non-technical users with an accessible method of interacting with RIGBY while maintaining access to core LeRobot functionality, including dataset recording and policy execution.

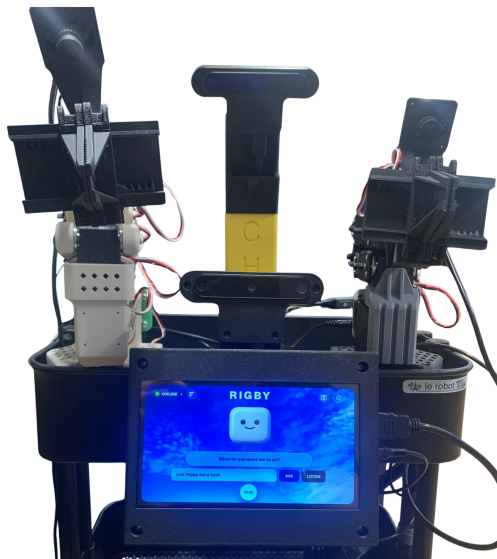


Figure 6.5: By presenting task selection, system feedback, and interaction controls through a single visual interface, the GUI reduces the need for specialist robotics knowledge and makes the system easier to access through familiar devices.

As RIGBY operates as a self-contained mobile platform, the user interaction system was designed to run directly on the Jetson. The interface can be displayed locally through an attached HDMI display while also being served as a web application accessible from any device connected

to the local network.

This allows the system to be operated from laptops, smartphones, AAC devices, and wheelchair-mounted tablets, helping to improve accessibility for users with limited mobility while maintaining the same functionality across all access methods.

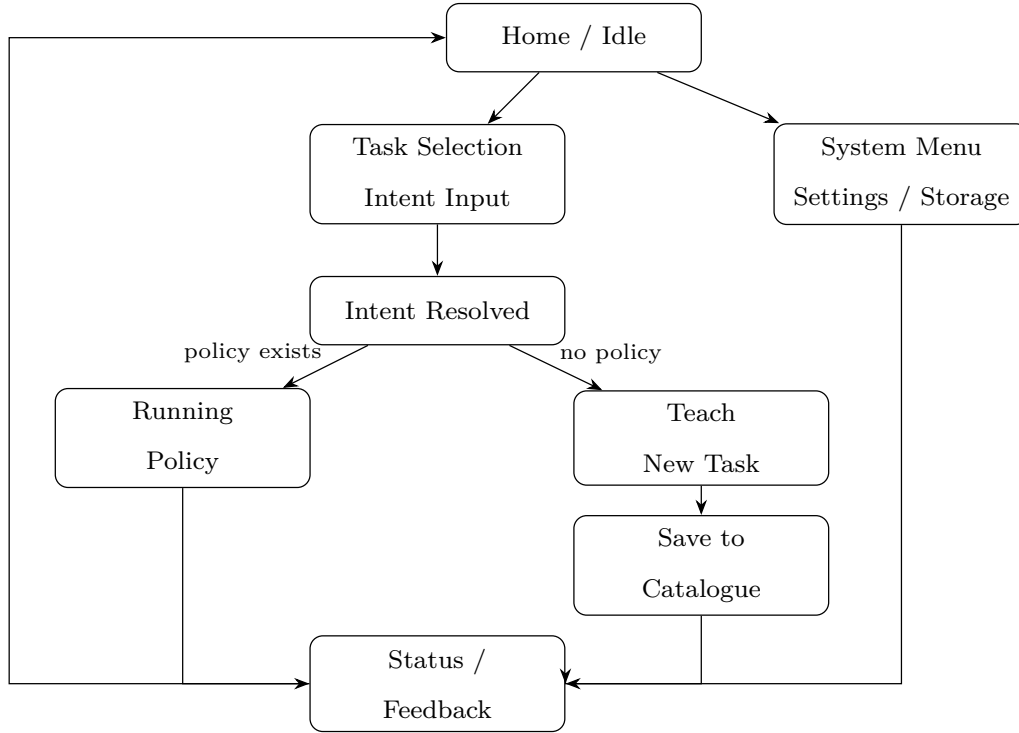


Figure 6.6: High-level GUI state model for RIGBY. The interface is centred around an idle home state, from which the user may enter task selection, policy execution, teaching, or system configuration workflows.

6.3.1 RIGBY User Interface Design

The RIGBY interface acts as a coordination layer over the existing software stack, providing a single point of access to the core functionality required throughout the imitation learning workflow. As shown in Figure 6.7, users can interact with the robot through either typed or spoken natural language commands while monitoring the current system status. The interface additionally provides direct access to demonstration recording, policy execution, and diagnostic tools, reducing reliance on command-line interaction.



Figure 6.7: The RIGBY home interface reduces operational complexity by combining task control, voice and text interaction, and system status feedback in a single accessible screen.

To further improve accessibility, RIGBY presents itself as an interactive character through a front-mounted touchscreen display. Facial expressions and visual state transitions provide continuous feedback regarding the robot’s internal state, as shown in Figure 6.8. This feedback was particularly important given the approximately 9.4-second delay observed between receiving a command and initiating policy execution, resulting from stages such as speech recognition, language model inference, task selection, and policy initialisation. By making these intermediate states visible, the interface reduces ambiguity and reassures users that requests have been received and are being processed.

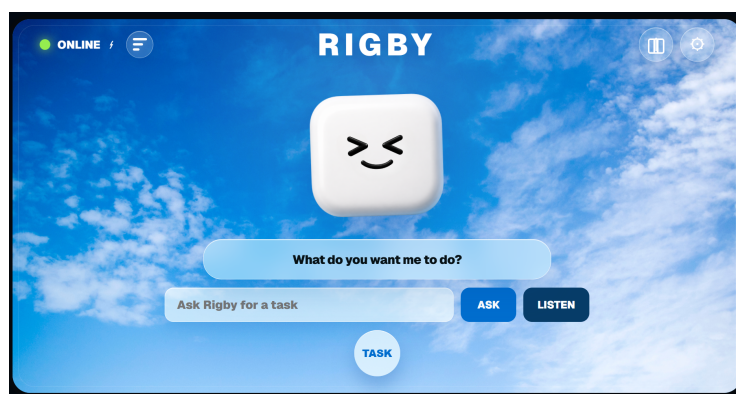


Figure 6.8: Facial expression changes make RIGBY’s interaction state more legible, supporting a more approachable human-robot interaction experience.

Direct interaction with the touchscreen reinforces the connection between the user and the physical platform while providing an additional channel for feedback and engagement.

6.3.2 Task Catalogue and Policy Binding

Rigby’s task execution framework is built around a fixed task catalogue that associates user-facing task descriptions with trained policy checkpoints. Although a local large language model is used to interpret natural language requests, it is not permitted to generate arbitrary robot actions. This design follows the principle of grounding natural language requests in available robot capabilities, as demonstrated by SayCan, where language-model outputs are constrained by feasible robot affordances [27]. Instead, its role is limited to mapping user intent onto predefined tasks within the catalogue.

This design treats natural language as an intent interface instead of a direct control mechanism. A policy is executed only if a corresponding task entry and valid checkpoint exist. Otherwise, the user can be directed towards collecting demonstrations and training a new policy for the requested capability.

Separating task selection from policy execution improves reliability and ensures that the system maintains awareness of whether it is currently recording demonstrations, configuring tasks, or executing an existing policy.

Perception-Assisted Task Discovery

RIGBY includes a perception-assisted task discovery mode to reduce reliance on manual policy selection. When requested, OAK-D camera frames are processed using DepthAI/OpenCV and a YOLOv8 detector from the Ultralytics Python library. Detected objects are matched against the fixed catalogue of trained policies so that RIGBY can suggest relevant behaviours.

Perception is therefore used only to support policy selection, not to generate robot actions, preserving the safety and predictability of validated policy execution.

The overall interaction framework combines natural language understanding, perception-assisted task discovery, fixed policy binding, and execution monitoring within a unified decision-making process. Instead of translating user requests directly into low-level robot actions, RIGBY interprets requests as high-level intents that are resolved to validated task policies where possible. When no suitable policy exists, the system can transition into a teaching workflow, enabling the robot’s capabilities to be extended through demonstration. The complete task-resolution procedure is summarised in Algorithm 3.

6.3.3 Dataset and Calibration Management

A dedicated interface was developed to simplify management of datasets and calibration files used throughout the imitation learning pipeline. As shown in Figure 6.9, operators can view, load, and remove calibration files and datasets through a graphical interface.

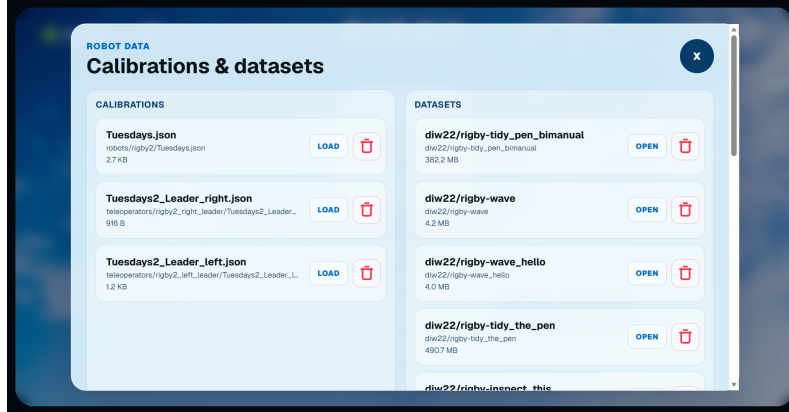


Figure 6.9: The dataset and calibration management interface lowers the barrier to policy training by making stored demonstrations and robot setup files visible and manageable.

This functionality removes the need to manually navigate the underlying filesystem and provides a consistent mechanism for managing training assets. The interface was designed to support rapid experimentation, allowing operators to switch between robot configurations and datasets without leaving the UI.

6.3.4 Demonstration Recording and Training Configuration

When a user requests a task for which no corresponding policy exists, RIGBY is able to transition from task execution into a learning workflow. As opposed to attempting to generate an unsafe or unpredictable response, the system informs the user that the requested capability is unavailable and guides them towards demonstrating the task. This interaction model allows new behaviours to be incorporated into the robot’s task catalogue through imitation learning while maintaining the safety benefits associated with fixed policy binding.

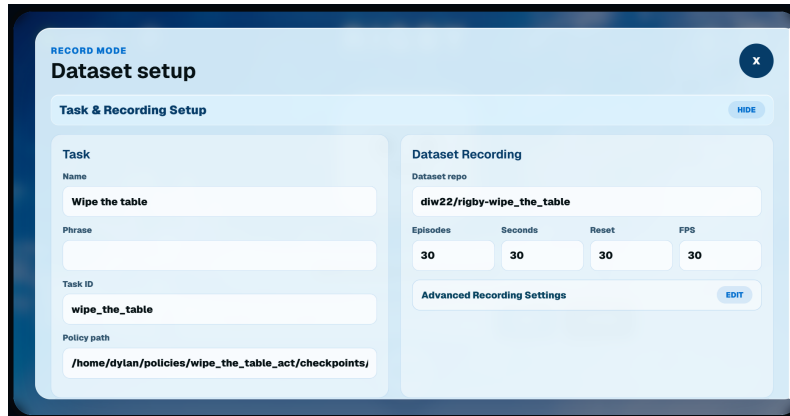


Figure 6.10: The recording and training interface turns demonstration collection and policy configuration into a guided workflow for non-expert users.

The demonstration collection workflow is exposed through a dedicated interface shown in Figure 6.10. Operators can configure task names, dataset repositories, recording parameters, and training settings from a single screen. By presenting these options through form-based controls, the interface reduces the complexity traditionally associated with dataset creation and policy training, lowering the barrier to extending the robot’s capabilities without requiring direct interaction with command-line tools.

6.3.5 System Inspection and Diagnostics

To improve system robustness and support troubleshooting, the interface includes dedicated diagnostic functionality. Figure 6.11 illustrates the object inspection interface, which provides live visual feedback from the robot’s perception pipeline.

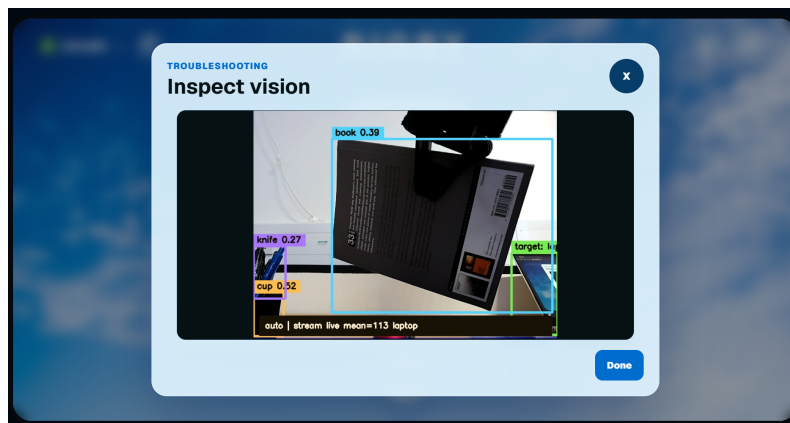


Figure 6.11: The object inspection interface connects perception diagnostics with task discovery, allowing recognised objects to support policy selection and future assistive behaviours.

Providing direct access to perception outputs allows operators to verify that cameras, object detectors, and inference pipelines are functioning correctly before task execution. This capability is particularly useful during development and demonstration collection, where many failures originate from perception errors instead of manipulation policies.

Collectively, these interfaces provide a unified environment for interaction, demonstration collection, policy deployment, system monitoring, and troubleshooting while remaining aligned with the project’s goals of accessibility, capability, and robustness.

6.3.6 User Interaction Modalities

RIGBY supports multiple interaction modalities to accommodate both development workflows and end-user operation. Alternative interaction pathways were retained to improve robustness during demonstrations and simplify testing of individual components within the interaction pipeline. The supported interaction modes are summarised in Table 6.1.

Table 6.1: Interaction modalities supported by the RIGBY interface.

Interaction Mode	Typical Use Case	Supported Functionality
Graphical interface	General operation	Demonstration recording, policy execution, task configuration, and system monitoring.
Text input	Development and debugging	Intent testing and validation of the task selection pipeline without requiring audio input.
Live voice commands	Natural interaction	Speech-driven task requests using local speech recognition and intent parsing.
Wake-word activation	Hands-free operation	Launching the RIGBY software stack and forwarding spoken commands to the backend.

Voice interactions follow a processing pipeline consisting of speech capture, speech-to-text transcription, intent interpretation, task selection, and graphical feedback. Transcription is performed locally using Whisper, while intent parsing is handled by a locally deployed Ollama model. Local inference was selected to avoid dependence on internet connectivity.

Algorithm 3 RIGBY Interaction and Task Coordination Procedure

```

1: procedure RESOLVEINTENT(userCommand)
2:   tasks  $\leftarrow$  LoadTaskCatalogue()
3:   if IsWakeCommand(userCommand) then
4:     DisplayWakeResponse()
5:     return
6:   end if
7:   if IsEndSessionCommand(userCommand) then
8:     DisplayEndSessionResponse()
9:     return
10:  end if
11:  if HasPendingSuggestion() then
12:    confirmation  $\leftarrow$  HandlePendingConfirmation(userCommand)
13:    if confirmation  $\neq \emptyset$  then
14:      return confirmation
15:    end if
16:  end if
17:  if ShouldRunObjectInspection(userCommand) then
18:    result  $\leftarrow$  RunInspectionFlow(userCommand)
19:    StoreSuggestedPolicy(result)
20:    return result
21:  end if
22:  taskId, taskConfig  $\leftarrow$  MatchTask(userCommand, tasks)
23:  if taskId =  $\emptyset$  then
24:    DisplayMessage("I don't know how to do that yet.")
25:    SetState(needs_teaching)
26:    return
27:  end if
28:  SetState(running_task)
29:  DisplayMessage("Okay, I will run this task.")
30:  result  $\leftarrow$  RunPolicy(taskId, taskConfig)
31:  if result.ok then
32:    SetState(idle)
33:    DisplayCompletionMessage()
34:  else
35:    SetState(error)
36:    DisplayErrorMessage(result)
37:  end if
38: end procedure

```

7

Testing Methodology

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This chapter describes the methodology used to evaluate the capabilities and limitations of the RIGBY platform. The evaluation was designed to test not only whether the robot could complete representative assistive manipulation tasks, but also how reliably and smoothly those behaviours could be executed. The selected tasks cover different forms of coordination, including sequential bimanual manipulation, object stabilisation, deformable-object interaction, and whole-body mobile manipulation.

7.1 Evaluation Tasks

To evaluate both teleoperation and autonomous execution, four manipulation tasks were selected spanning sequential manipulation, constrained object handling, deformable object interaction, coordinated bimanual manipulation, and whole-environment modification. Following the evaluation methodology adopted by Mobile ALOHA, manipulated objects were randomised within a 2 cm radius of their nominal pose before each trial [38]. Success was measured both at the task level and through intermediate milestones corresponding to key stages of task completion.

7.1.1 Pen Placement into a Pen Holder

The robot must place a pen into a desktop pen holder, as shown in Figure 7.1. The pen initially lies outside the robot workspace on top of a book. To complete the task, the robot must first slide the book towards itself before grasping the pen and inserting it into the holder.

The pen position was randomised within a 2 cm radius before each trial.



Figure 7.1: The pen placement task tests sequential support-dominant coordination, requiring the support arm to reposition the book before the dominant grasps and inserts the pen.

Milestones:

1. Book successfully repositioned.
2. Pen successfully grasped.
3. Pen successfully inserted into the holder.

A trial was considered successful when all milestones were completed.

7.1.2 Wine Glass Lift and Surface Wiping

The robot must lift a wine glass, wipe the table surface beneath it, and return the glass to the table, as shown in Figure 7.2.

The glass position was randomised within a 2 cm radius before each trial.

Milestones:



Figure 7.2: The wine glass lifting and wiping task tests parallel bimanual role allocation, with one arm maintaining object clearance while the other performs surface wiping.

1. Glass successfully grasped and lifted.
2. Surface beneath the glass successfully wiped.
3. Glass returned to the table.

A trial was considered successful when all milestones were completed without dropping the glass.

7.1.3 Chair Repositioning

The robot must reposition a lightweight chair by pushing it from an initial pose to a designated target region, as shown in Figure 7.3. The task requires coordinated operation of the manipulator and mobile base, with the arm maintaining contact with the chair while the robot navigates around the environment.

Prior to each trial, the chair position and orientation were randomised within a 2 cm translational radius and $\pm 5^\circ$ rotational offset from their nominal configuration.

Milestones:

1. Successful establishment of contact between the manipulator and chair.
2. Chair displacement initiated in the intended direction.
3. Chair successfully repositioned within the target region.



Figure 7.3: The chair repositioning task tests whole-body manipulation, requiring the robot to sustain contact and use mobile-base motion to move an external object into a target region.

A trial was considered successful when the chair came to rest with its base contained within the target region.

7.1.4 Ziploc Bag Closure

The robot must close the ziploc mechanism along the full length of the bag opening, as shown in Figure 7.4.

The bag position was randomised within a 2 cm radius before each trial.

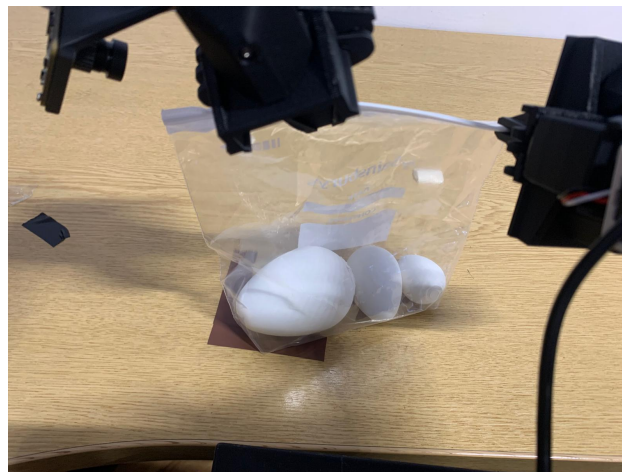


Figure 7.4: The Ziploc bag closure task tests deformable-object bimanual coordination, requiring one arm to stabilise the bag while the other performs the closure motion.

Milestones:

1. Zipper successfully engaged.
2. Zipper motion initiated.
3. Bag successfully sealed.

A bag was considered sealed when more than 90% of the zipper length was closed.

7.2 Motion Quality Evaluation

In addition to task success rate and completion time, motion quality was evaluated by measuring the smoothness of the executed action trajectories. The objective was to determine whether the learned policy produced smooth control commands during autonomous execution. Lower jerk values have been widely associated with smoother and more natural movement patterns [39].

LeRobot records the action vector applied at each timestep. For each episode, the recorded action sequence was extracted as

$$a(t) = \begin{bmatrix} a_1(t) & a_2(t) & \cdots & a_D(t) \end{bmatrix}^T,$$

where D is the dimensionality of the robot action space. In this work, the action vector contains both arm and mobile base command dimensions, and the resulting metric is therefore interpreted as a measure of command-space smoothness.

Jerk was defined as the third derivative of the action trajectory with respect to time,

$$j_a(t) = \frac{d^3 a(t)}{dt^3}.$$

Since actions were recorded discretely at sampling frequency f_s , jerk was estimated using the third finite difference,

$$j_a[k] \approx \Delta^3 a[k] f_s^3,$$

where

$$\Delta^3 a[k] = a[k+3] - 3a[k+2] + 3a[k+1] - a[k].$$

For each episode, RMS action jerk was computed as

$$J_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{k=1}^N \|j_a[k]\|_2^2},$$

where N is the number of valid jerk samples in the episode. Lower RMS jerk values correspond to smoother action trajectories.

RMS action jerk was computed for each autonomous evaluation episode and averaged across successful trials for each task. This metric does not measure physical end-effector jerk directly, since no forward kinematic reconstruction was applied. Instead, it provides a consistent measure of smoothness in the policy command space, allowing different policies to be compared under the same robot action representation.

7.3 Reported Metrics

For each task, the following metrics were reported:

- Milestone completion rate (%)
- Overall task success rate (%)
- Average task completion time
- RMS action jerk

Task completion time provides an indication of the practical usability of the learned policies, while RMS action jerk was used to assess the smoothness of the generated control commands. Together, these metrics provide insight into both the effectiveness and quality of autonomous task execution.

8

Results and Analysis

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This chapter presents the experimental results obtained from evaluating ACT and DOM-ACT on the selected manipulation tasks. The analysis first considers whether both policies trained successfully, before comparing task success, milestone completion, completion time, and motion smoothness. The final part of the chapter examines system-level performance during deployment to determine whether computational constraints contributed to the observed motion quality.

8.1 Learning Dynamics

To assess whether differences in downstream task performance could be attributed to optimisation behaviour, the training losses of the ACT and DOM-ACT policies were compared during policy learning. Figure 8.1 shows the training-loss curves for the pen placement policy over a matched training horizon of 15,000 steps.

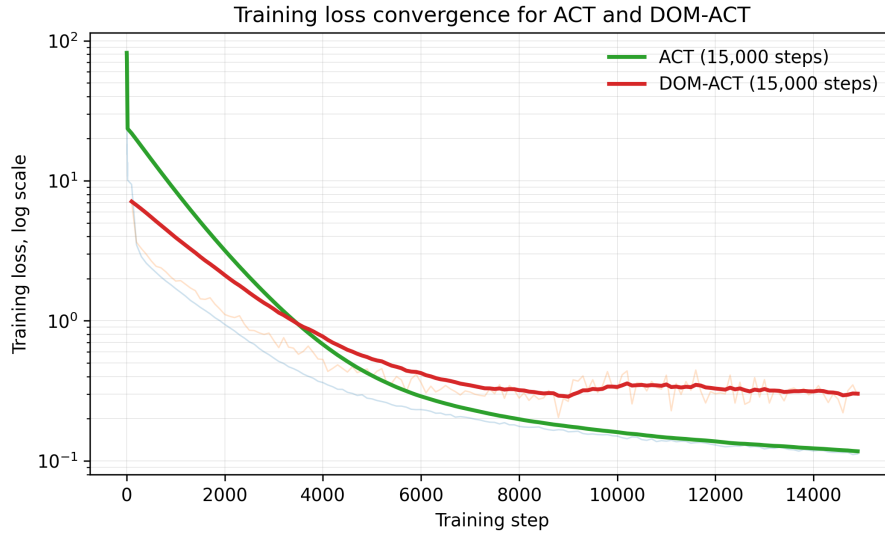


Figure 8.1: Training-loss curves for ACT and DOM-ACT on the pen placement policy. Both architectures indicate stable optimisation and successful learning.

As shown in Figure 8.1, both policies exhibit successful learning dynamics, with training loss decreasing substantially over the course of optimisation. When compared over the same 15,000-step training horizon, both ACT and DOM-ACT show a clear downward trend followed by convergence toward a lower-loss regime. This indicates that both architectures were able to optimise successfully, instead of suffering from instability or training collapse.

This result is important for interpreting the task-performance findings. Since both architectures train successfully, the differences observed in autonomous task success are unlikely to arise from one model failing to learn.

8.2 Task Performance

Task performance was evaluated using milestone completion rates and overall task success rates. Milestone-based evaluation was used because several tasks involve multiple sequential stages, where partial completion provides useful information about the source of failure. This allows differences between ACT and DOM-ACT to be analysed not only in terms of final success, but also in terms of which subtasks were most affected by the policy architecture.

Table 8.1: Milestone completion rates for the pen placement task. Values are reported over 20 evaluation trials.

Milestone	ACT	DOM-ACT
Book repositioned (%)	90	100
Pen grasped (%)	70	80
Pen inserted into holder (%)	65	80
Overall task success (%)	65	80

Table 8.2: Milestone completion rates for the wine glass lifting and wiping task.

Milestone	ACT	DOM-ACT
Wine glass grasped (%)	95	100
Wine glass lifted (%)	95	100
Towel grasped and surface wiped (%)	95	85
Wine glass returned (%)	95	100
Overall task success (%)	95	85

Table 8.3: Milestone completion rates for the ziploc closure task.

Milestone	ACT	DOM-ACT
Bag successfully grasped (%)	55	65
Zipper engagement achieved (%)	35	45
Closure motion initiated (%)	35	45
Bag fully closed (%)	35	40
Overall task success (%)	35	40

Table 8.4: Milestone completion rates for the chair repositioning task. Values are reported over 20 evaluation trials.

Milestone	ACT	DOM-ACT
Contact established (%)	100	100
Chair displacement initiated (%)	100	100
Chair repositioned within target region (%)	100	100
Overall task success (%)	100	100

8.2.1 Discussion

The results suggest that the effectiveness of DOM-ACT depends strongly on whether the dominant-support assumption aligns with the coordination requirements of the task.

For the pen placement task, DOM-ACT improved overall task success from 65% to 80% (Table 8.1). Improvements were observed across all milestones, with book repositioning increasing from 90% to 100%, pen grasping from 70% to 80%, and successful pen insertion from 65% to 80%. Importantly, successful completion of the dominant arm objective (grasping and inserting the pen) relied upon prior completion of the supporting arm action of repositioning the book. The observed improvements therefore may suggest that explicitly modelling dominant-support relationships may be beneficial when one manipulator prepares the environment for the subsequent actions of the other.

A similar trend was observed for the ziploc closing task, where DOM-ACT improved overall success from 35% to 40% (Table 8.3). Incremental improvements were evident across all milestones, including bag grasping, zipper engagement, and initiation of the closing motion. Although the overall performance remained relatively low due to the intrinsic difficulty of the task, the consistent gains suggest that the dominant-support inductive bias may provide advantages in tasks involving sequential dependencies between the two manipulators.

By contrast, the wine glass lifting and wiping task produced the opposite effect. While DOM-ACT improved the success rates associated with the left-arm subtasks, increasing successful wine glass grasping and lifting from 95% to 100%, performance on the wiping milestone decreased from 95% to 85% (Table 8.2). Consequently, overall task success fell from 95% to 85%. In this case, successful completion of the wiping behaviour depended upon accurate execution of the arm

responsible for stabilising the wine glass, indicating a more reciprocal coordination structure than assumed by the fixed dominant-support decomposition.

The chair repositioning task achieved a 100% success rate across all milestones for both ACT and DOM-ACT (Table 8.4). Unlike the other evaluated tasks, chair repositioning primarily required coordinated mobile manipulation as opposed to complex bimanual interaction. While these results demonstrate that both policies were capable of integrating arm and base behaviours to accomplish the task robustly, the task did not provide sufficient difficulty to distinguish between the two architectures.

These findings suggest that the inductive bias introduced by DOM-ACT is neither universally beneficial nor universally detrimental. Rather, its effectiveness appears to depend on the extent to which the assumed dominant-support relationship reflects the true dependency structure of the task. Tasks exhibiting clear sequential dependencies may benefit from explicit role separation, whereas tasks requiring tightly coupled or bidirectional coordination may require more flexible mechanisms capable of dynamically allocating dominant and supporting roles.

8.2.2 Quality

In addition to task success rates, the quality of autonomous task execution was evaluated using two complementary metrics: task completion time and trajectory smoothness. Completion time provides an indication of the practical usability of the learned policies, while smoothness reflects the quality of the generated motions.

As previously mentioned, trajectory smoothness was quantified using the Root Mean Square (RMS) jerk of the end-effector trajectory. Lower RMS jerk values correspond to smoother motions and have previously been associated with more natural and acceptable robot behaviour.

Table 8.5: Task quality metrics without temporal ensembling. Lower RMS jerk values indicate smoother trajectories.

Task	Completion Time (s)		RMS Jerk	
	ACT	DOM-ACT	ACT	DOM-ACT
Pen placement	23.5	24.2	11,300	10,900
Wine glass wiping	15.2	17.3	9,800	10,100
Ziploc closing	23.4	24.1	14,500	14,000
Chair repositioning	18.7	18.4	8,000	7,500

As shown in Table 8.5, RMS jerk values were broadly comparable between ACT and DOM-ACT across all evaluated tasks. Similarly, completion times differed only marginally between policies. These findings suggest that, despite differences in task success rates, the proposed dominant-support decomposition did not substantially influence the quality of the resulting motions. Instead, both architectures appeared to generate trajectories of comparable smoothness and efficiency during autonomous execution.

Given the comparable motion quality observed between ACT and DOM-ACT, subsequent analysis investigated whether temporal ensembling could improve trajectory smoothness independent of policy architecture.

Table 8.6: Effect of temporal ensembling (TE) on task quality metrics. Values represent averages across ACT and DOM-ACT evaluations. Lower RMS jerk values indicate smoother trajectories.

Task	Completion Time (s)		RMS Jerk	
	No TE	TE	No TE	TE
Pen placement	23.9	61.2	11,100	9,900
Wine glass wiping	16.3	34.8	9,950	8,400
Ziploc closing	23.8	58.5	14,250	13,500
Chair repositioning	18.6	39.1	7,750	7,000

Table 8.6 shows that temporal ensembling consistently reduced RMS jerk across all evaluated tasks, indicating modest improvements in motion smoothness. However, these improvements came at the expense of substantially increased execution times. For the pen placement and ziploc closing tasks, completion times increased by approximately a factor of three, while the wine glass wiping

and chair repositioning tasks exhibited approximately two-fold increases in duration.

Consequently, although temporal ensembling improved motion quality, the magnitude of the improvement was relatively small compared to the associated reduction in execution efficiency. This trade-off may limit the practicality of temporal ensembling for assistive applications, where timely task completion is often as important as smooth execution.

Despite the use of temporal ensembling, RMS jerk values remained substantially higher than those observed during teleoperated demonstrations. Given that temporal ensembling explicitly aims to smooth action sequences, this suggests that factors beyond the learned policy architecture contributed to the reduced motion quality. This suggests that factors beyond the learned policy architecture contributed to the reduced motion quality.

To investigate this further, the execution frequency of the LeRobot evaluation loop was monitored during autonomous operation. As shown in Figure 8.2, the realised control frequency periodically fell below the intended operating frequency of 15Hz.

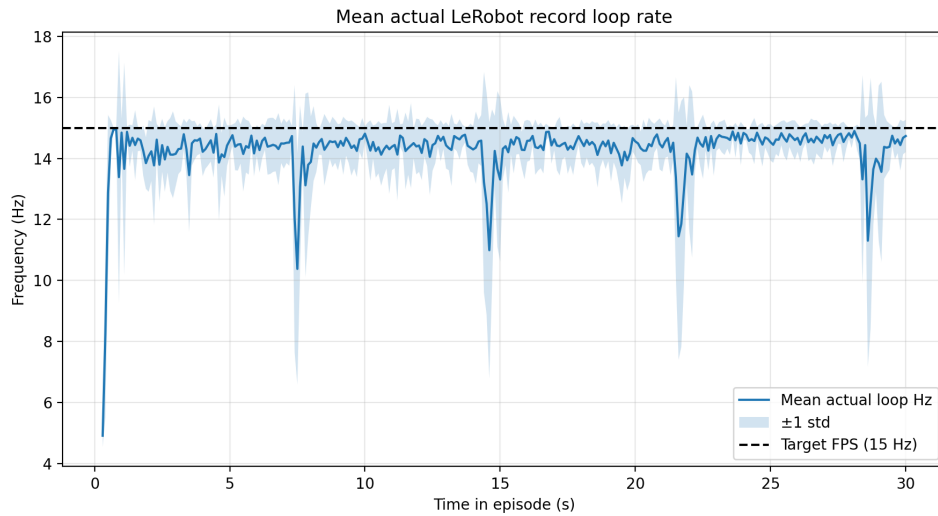


Figure 8.2: The LeRobot execution-frequency trace shows that policy deployment periodically falls below the 15 Hz target, indicating real-time constraints from computation and system overhead.

Concurrent CPU profiling revealed substantial computational demands arising from policy execution, recording, and video processing. Figure 8.3 shows that the `lerobot-record` process progressively consumed in excess of 100% CPU utilisation, while `ffmpeg` consistently occupied approximately 50% CPU utilisation throughout execution.

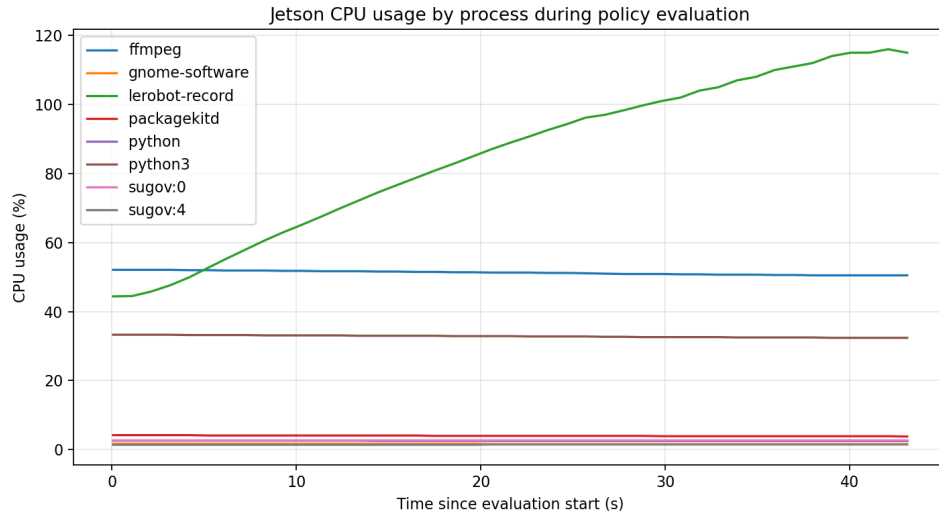


Figure 8.3: CPU utilisation during policy evaluation identifies increasing lerobot-record load and sustained ffmpeg demand as likely contributors to execution-frequency drops.

Since ACT-based policies assume approximately fixed temporal intervals between successive actions, variations in execution frequency may result in uneven actuator updates and consequently increased jerk. The elevated jerk values observed in autonomous execution therefore likely reflect not only characteristics of the learned policies, but also limitations imposed by the computational resources available on the embedded deployment platform. These findings highlight the importance of considering system-level factors when evaluating imitation learning policies on resource-constrained robotic systems.

Future work should therefore investigate real-time scheduling strategies, dedicated inference pipelines, and hardware acceleration techniques to minimise execution variability and better realise the smooth motions demonstrated during teleoperation.

Overall, the results suggest that incorporating task-relevant inductive biases into bimanual imitation learning may improve performance for tasks exhibiting clear dominant-support relationships. However, the findings also demonstrate that policy performance in embodied systems cannot be considered independently from the computational infrastructure on which they are deployed. Execution frequency variability arising from resource limitations on low-cost hardware may substantially influence the smoothness and reliability of learned behaviours.

9

Evaluation

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This chapter evaluates the outcomes of the project in relation to the original objectives and the broader goals of accessible assistive robotics. While Chapter 8 presented the experimental results, this chapter interprets their significance at the system level. It considers whether RIGBY achieved its design aims, how it compares with existing robotic platforms, and what limitations remain before low-cost mobile bimanual manipulators can be considered practical assistive technologies.

9.1 Evaluation of Design Objectives

The primary objective of this project was to investigate whether a low-cost mobile bimanual manipulator could support meaningful assistive activities of daily living through demonstration-based learning. This objective was largely achieved. RIGBY integrated mobile manipulation, RGB-D sensing, whole-body teleoperation, demonstration recording, imitation learning, and autonomous policy execution within a platform costing approximately £1951. The system was able to perform a range of representative assistive manipulation tasks, demonstrating that useful mobile bimanual behaviours can be investigated without relying on prohibitively expensive research hardware.

A second objective was to improve accessibility by reducing reliance on specialist robotics

expertise. This was addressed through the multimodal interface, which supports various interaction modes, alongside tools for policy training and deployment. These features reduced the need for users to interact directly with command-line tools or low-level configuration files. However, as formal user studies were beyond the scope of this work, usability claims should be interpreted as design-led outcomes, not empirically validated accessibility results.

The final objective was to investigate whether dominant-support inductive biases could improve bimanual imitation learning. This objective was partially achieved. DOM-ACT improved performance on tasks with clear sequential dependencies between manipulators, but did not consistently improve tasks requiring more reciprocal coordination. This suggests that dominant-support decomposition may be a useful task-specific bias.

Overall, the design objectives were substantially met. RIGBY demonstrated the feasibility of low-cost mobile bimanual manipulation while revealing limitations in robustness, motion quality, computation, and generalisation that must be addressed before such systems can be considered practical assistive technologies.

9.2 Evaluation of the Implementation Approach

The project attempted to develop a complete embodied system combining low-cost hardware, teleoperation, imitation learning, user interaction, and autonomous policy execution. This was appropriate given the aim of investigating whether an accessible mobile bimanual assistive platform could be built and evaluated as an integrated system.

A strength of this approach was that it exposed practical constraints that would not have been visible from simulation or policy evaluation alone. Issues such as actuator limitations, control-frequency variation, calibration sensitivity, demonstration quality, and onboard computational load all influenced the final robot behaviour. This made the evaluation more representative of the challenges involved in deploying learning-based control on low-cost physical hardware.

However, the engineering-led approach also introduced risks. Several stages of the project were delayed by hardware failures, redesigns, and reliability issues. In particular, the development of the custom manipulator would have benefited from more detailed analysis before fabrication, such as estimating joint loading, assessing expected stresses, and validating the design through simulation or simplified mechanical calculations. Performing this analysis earlier may have reduced the number of physical iterations required.

The project therefore demonstrates both the value and difficulty of developing low-cost embodied robotic systems. Rapid prototyping enabled fast progress and practical learning, but greater use of upfront mechanical analysis, modular subsystem testing, and staged validation would have reduced risk during integration. Future development of RIGBY should therefore adopt a more formal design-validation process before major hardware changes are manufactured or deployed.

9.3 Critical Evaluation of RIGBY

Table 9.1: Comparison between RIGBY and representative research platforms. Costs are approximate and intended only to illustrate relative differences in accessibility and capability.

Platform	Approx. Cost	Bimanual	Mobile	Imitation Learning
PR2 [40]	>£250,000	✓	✓	Limited
YuMi [25]	~£40,000	✓	×	Possible
Mobile ALOHA [19]	~£25,000	✓	✓	✓
RIGBY (this work)	£1,951	✓	✓	✓

As shown in Table 9.1, RIGBY should not be interpreted as a direct replacement for established research platforms such as PR2, YuMi, or Mobile ALOHA. Those systems provide greater robustness, payload capacity, actuation quality, and manipulation performance. RIGBY instead occupies a different position within the design space, prioritising affordability, reproducibility, and accessibility while retaining enough capability to investigate assistive mobile manipulation.

The results therefore represent a trade-off, not an unqualified success. The system completed meaningful assistive behaviours, but performance remained constrained by low-cost actuators, limited onboard computation, and environment-specific imitation learning. In particular, execution profiling indicated that control frequency periodically fell below the intended operating rate, which likely contributed to increased jerk and reduced motion quality during autonomous execution. This highlights that embodied learning systems cannot be evaluated purely as policies or algorithms; the physical and computational platform strongly influences the quality of the final behaviour.

The motion quality results also show that task success alone is not sufficient for evaluating assistive robots. ACT and DOM-ACT produced broadly comparable completion times and trajectory smoothness, while temporal ensembling only modestly reduced RMS jerk at the expense of substantially longer task durations. This illustrates a practical trade-off between smoothness and

responsiveness. In assistive settings, smoother motion may improve perceived safety and comfort, but overly slow execution may reduce usefulness.

The comparison between ACT and DOM-ACT further demonstrates the importance of evaluating the assumptions embedded within learning architectures. DOM-ACT was beneficial when the task structure matched its dominant-support formulation, but less effective when coordination between the two arms was more balanced. However, these results should not be interpreted as isolating the dominant-support inductive bias alone, since DOM-ACT also introduces separate dominant and support decoders, dominant-arm conditioning of support and base predictions, partitioned action losses, additional regularisation terms, and potentially a different parameter count. Future ablation studies and parameter-matched comparisons would therefore be required to determine whether the observed improvements arise from the intended inductive bias, increased model capacity, the modified optimisation objective, or other architectural differences between the models.

Overall, an appropriate benchmark for platforms such as RIGBY is not whether they outperform premium research systems, but whether they provide sufficient capability to broaden access to assistive robotics research. From this perspective, the project was successful. RIGBY demonstrated mobile bimanual manipulation, learning from demonstration, whole-body teleoperation, and multimodal interaction on a low-cost platform, while also exposing the practical constraints that must be addressed for real-world deployment.

9.4 Limitations

Several limitations should be recognised when interpreting the results and the evaluation of the proposed approach. First, the experimental evaluation was conducted using a limited set of structured tasks in controlled environments. Although these tasks represented meaningful assistive behaviours, they do not capture the variability of real domestic settings, where object placement, clutter, lighting, and user behaviour are less predictable.

Second, the learned policies were strongly tied to the demonstrations and environments used during training. Object position, camera viewpoint, lighting, and initial robot configuration remained important factors in successful execution. This limits the extent to which the system can currently generalise to unseen homes, objects, or users.

Third, the hardware introduced limitations in repeatability and control quality. The use of

low-cost actuators reduced the financial barrier to entry, but also increased sensitivity to backlash, compliance, calibration error, and load variation. These effects can accumulate during bimanual manipulation and contribute to failures that are not solely caused by the learning algorithm.

Fourth, the evaluation did not include formal studies with end users. Although the interface was designed to improve accessibility, factors such as trust, perceived safety, comfort, predictability, and ease of use require direct assessment with representative users before stronger claims can be made about practical assistive value.

Finally, DOM-ACT assumes a fixed dominant-support relationship between the manipulators. The results suggest that this assumption is not appropriate for all bimanual tasks. Future architectures should therefore consider adaptive role allocation mechanisms that can change the coordination structure during execution.

10

Conclusions

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10.1 Conclusion

This dissertation presented the design, development, and evaluation of RIGBY, a low-cost mobile bimanual manipulator for assistive activities of daily living. The system combined mobile manipulation, RGB-D sensing, whole-body teleoperation, imitation learning, and multimodal user interaction within a hardware platform costing approximately £1951.

Experimental evaluation showed that RIGBY was capable of learning and executing a range of representative assistive manipulation tasks. These results demonstrate that meaningful mobile bimanual manipulation research can be conducted using substantially lower-cost hardware than many established research platforms. However, the results also show that affordability introduces important constraints. Motion quality, control frequency, actuator reliability, and task generalisation were all affected by the limitations of the low-cost system.

The comparison between ACT and DOM-ACT demonstrated that task structure plays an important role in bimanual imitation learning. DOM-ACT provided benefits for tasks with clear dominant-support dependencies, but did not consistently improve performance across all tasks.

This suggests that fixed coordination assumptions are insufficient for the full range of assistive manipulation scenarios, and that future systems may require more adaptive mechanisms for assigning roles between manipulators.

Overall, this work contributes an accessible platform for investigating assistive mobile bimanual manipulation, while also highlighting the practical challenges involved in deploying learning-based control on resource-constrained robotic systems. The value of low-cost platforms lies not only in reducing financial barriers, but also in revealing the sensing, computation, control, and reliability challenges that must be addressed if assistive robots are to move beyond laboratory demonstrations and become practical technologies for everyday environments.

10.2 Future Work

The findings of this dissertation suggest several directions for future research. A first direction is to use the URDF model developed in this project to support simulation-based demonstration collection. Collecting demonstrations directly on physical hardware is time-consuming, limits dataset diversity, and contributes to mechanical wear. Future work could investigate generating digital twins of domestic environments and using the RIGBY model within simulation to collect larger and more varied demonstration datasets. These demonstrations could then be transferred to the physical robot using sim-to-real methods [41]. This approach may improve robustness while reducing dependence on the physical platform during data collection.

A second direction concerns the computational infrastructure used for policy deployment. Profiling indicated that control frequency periodically dropped below the intended operating rate, which likely contributed to reduced motion quality. Future work should therefore evaluate alternative embedded computers, dedicated inference accelerators, and real-time scheduling strategies. This would help identify the minimum practical computing requirements for low-cost assistive manipulation without undermining the affordability of the platform.

Future evaluation should also include a wider range of assistive tasks and more realistic domestic conditions. The tasks used in this work provided useful benchmarks for coordinated manipulation, but real homes involve greater variation in object appearance, placement, clutter, lighting, and user preference. Studies involving representative users would also be required to evaluate trust, perceived safety, predictability, comfort, and practical usefulness.

Finally, the task-dependent performance of DOM-ACT suggests that future work should inves-

tigate adaptive bimanual coordination. Instead of assuming a fixed dominant and supporting arm, future policies could learn when to assign, exchange, or share roles between manipulators during execution. Such approaches may preserve the benefits of explicit role structure while providing the flexibility required for more complex activities of daily living.

Together, these directions point towards more robust, scalable, and user-centred low-cost assistive robots. Advances in simulation, embedded computation, adaptive coordination, and human-centred evaluation could help bridge the gap between accessible research prototypes and assistive robotic systems suitable for real-world deployment.



Bill of Materials

The complete bill of materials used throughout the development of RIGBY is provided in this appendix. Costs reflect actual expenditure where records were available and approximate retail prices otherwise. The purpose of including this information is to demonstrate the feasibility of constructing a research-grade mobile manipulation platform using predominantly off-the-shelf components at a substantially lower cost than comparable commercial systems.

Table A.1: Bill of materials for the RIGBY mobile manipulation platform. Costs reflect actual expenditure where records were available and approximate retail prices otherwise.

Component		Qty.	Unit	Cost	Total (£)	Purpose
			(£)			
<i>Actuation</i>						
FEETECH STS3215	servos (robot platform) [42]	17	19.71		335.00	Follower arms and mobile base
FEETECH STS3215	servos (teleoperation) [42]	14	19.75		276.50	Leader arms
Actuation subtotal					611.50	
<i>Computation and Interfaces</i>						
NVIDIA Jetson Orin Nano	Developer Kit [43]	1	187.50		187.50	Onboard computation
7-inch touchscreen display [44]		1	29.00		29.00	User interface
Logitech G Flight Simulator	Rudder Pedals [45]	1	100.00		100.00	Mobile base teleoperation
Powered USB hub [46]		1	50.00		50.00	Peripheral connectivity
Computation and interfaces subtotal					366.50	
<i>Perception</i>						
OAK-D context camera [47]		2	292.00		584.00	Environmental perception
Innomaker UCAM1080P [48]		2	20.00		40.00	Manipulation observation
Perception subtotal					624.00	
<i>Power and Integration</i>						
Anker SOLIX C300 [49]		1	200.00		200.00	Portable power supply
IKEA RÅSKOG trolley [50]		1	29.00		29.00	Structural frame
USB cables, adapters and wiring		N/A	N/A		120.00	System integration
Power and integration subtotal					349.00	
Estimated total platform cost					1951.00	

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B

User Guide

Full source code for the Rigby platform can be found here:

`https://github.com/diw22/FinalYearProject`

The repository includes instructions and models for assembling the hardware required to operate RIGBY. The repository also includes instructions for downloading and configuring the software required to operate RIGBY. The software components developed throughout this project include the user interface, teleoperation framework, intent-processing pipeline, policy execution system, and simulation assets.

The repository additionally contains the robot description files, task configuration files, and supporting utilities required to reproduce the experiments presented in this dissertation. Where permitted, trained policy checkpoints and demonstration datasets are also provided.

Further details regarding system setup, extending the task catalogue, and integrating additional hardware components can be found within the accompanying documentation.

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